Data S2: In depth characterization of 21 human OR for signature odorants summarized in main manuscript Figure 3

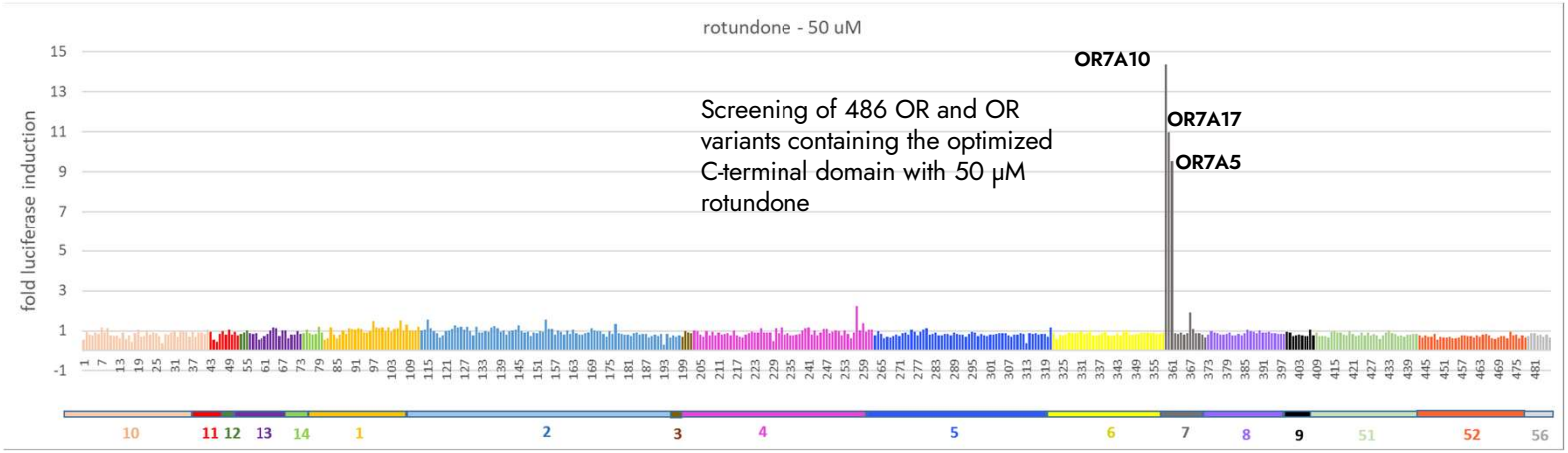
For each odorant of interest (or a simple mixture of related odorants), the range of all 486 OR and OR-variants with the optimized C-terminal sequences (Data file 1) was screened at one screening concentration.
Active OR where then tested on a library of 438 single fragrance ingredients to identify additional ligands. In parallel, structural analogues or analogues with similar odor description were selected from a larger odorant library with recorded odor descriptors and information on potency.

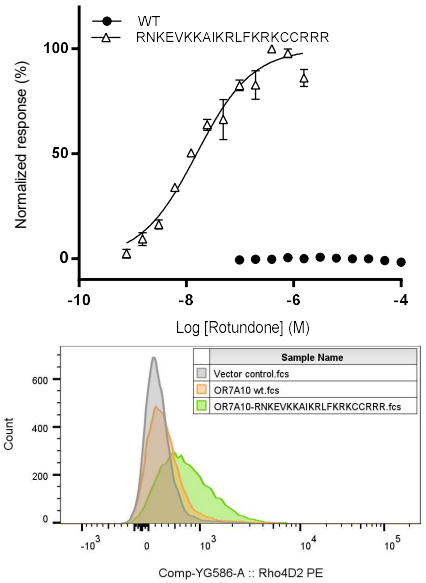
The potential ligands were investigated with dose-response analysis to establish structure-activity relationships. OR activation data are expressed as EC20% values and they are then compared to historical sensory data, namely subjective odor descriptions and objective measures of odor strength expressed as GTH and OTH. For those OR for which both the wild-type and the engineered variants were tested on either functional or surface expression, the improvement in functional / surface expression is also shown. For the SAR studies and de-orphanisation screens, only the optimized engineered variants were used.

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OR7A10 – the key OR for the pepper note rotundone





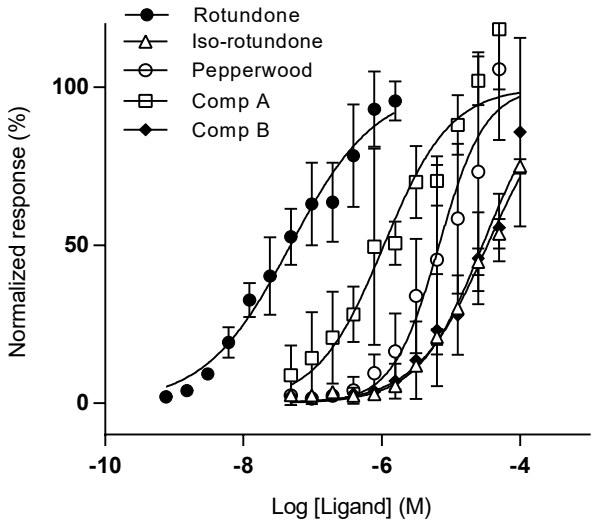
Activation and cell surface expression of OR7A10: Wild-type vs. optimized C-terminal sequence

Rotundone is a key peppery-woody impact odorant in pepper [1], but it is also an important trace odorant in a number of essential oils[2-3]. It imparts peppery notes to e.g. Syrah wines[1] and it is known for its exceptionally low detection threshold, making it a key impact odorant.

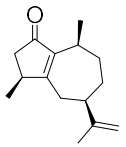
Rotundone activated OR7A10, OR7A5 and the woody OR7A17 in the pan-OR screening. OR7A10 can only be expressed with the modified C-terminus and was never described before. It is activated in the nanomolar range by rotundone, mirroring the low OTH/GTH.

The shift of the double bond in iso-rotundone leads to 1000-fold lower potency along with > 1000 fold higher OTH/GTH.

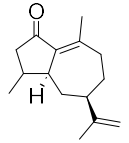
Among 438 fragrance chemicals, only pepperwood and patchouli oil, which contains 0.1% rotundone, activated OR7A10. Pepperwood is a commercial compound with similar odor description but very different structure, activating OR7A10 with a lower potency reflecting the higher OTH/GTH. The synthetic Oxadecalinones A and B [4] have similar odor descriptions, too, and indeed activate OR7A10. Their potency differs by a factor of 30 both *in vivo* and on OR7A10, hence the odor direction and relative strength of these analogues validate OR7A10 as the key OR for the perception of the peppery-woody note of rotundone.



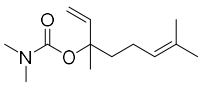
Activation of OR7A10 by peppery-woody odorants



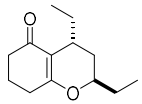
rotundone
woody, spicy, peppery guaïac wood
GTH: 0.0016 ng
OTH: 0.0004 ng/L
EC20% OR7A10: 6.7 x 10^{-9} M



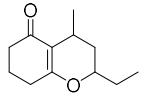
iso-rotundone
green, woody
GTH: 7.9 ng
EC20% OR7A10: 6.5 x 10^{-6}



Pepperwood
pepper, floral, woody
GTH: 4,1 ng
OTH: 2.7 ng/L
EC20% OR7A10: 2.3 x 10^{-6}



Comp A: (2*R*,4*R*)-2,4-diethyl-2,3,4,6,7,8-hexahydro-5*H*-chromen-5-one
woody, guaiacwood
GTH: 0.014 ng
EC20% OR7A10: 2.3 x 10^{-7} M

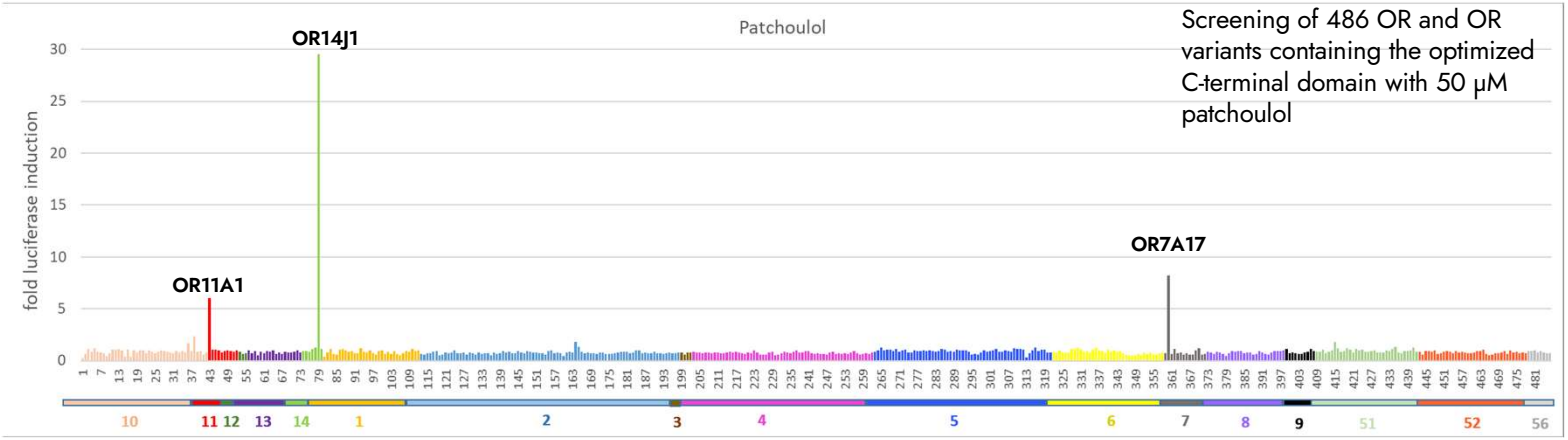


Comp B: 2-ethyl-4-methyl-2,3,4,6,7,8-hexahydro-5*H*-chromen-5-one
spicy(peppery), woody(guaïyl acetate)
GTH: 0.4 ng
OTH: 0.66 ng/L
EC20% OR7A10: 6.6 x 10^{-6} M

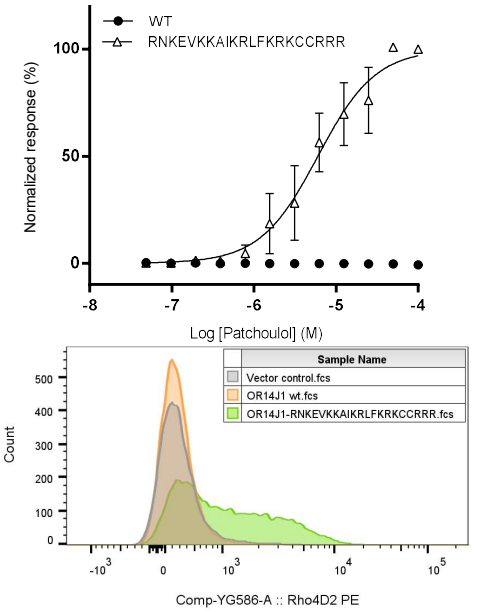
Structure, sensory data, and OR activation of ligands tested on OR7A10

1) Wood, C., Siebert, T. E., Parker, M., Capone, D. L., Elsey, G. M., Pollnitz, A. P., Eggers, M., Meier, M., Vössing, T., Widder, S., Krammer, G., Sefton, M. A.and Herderich, M. J., (2008) From wine to pepper: Rotundone, an obscure sesquiterpene, is a potent spicy aroma compound. J. Agric. Food Chem. 56, 3738-3744.10.1021/jf800183k.
2) Ishihara, M., Tsuneya, T.and Uneyama, K., (1991) Guaiane sesquiterpenes from agarwood. Phytochemistry 30, 3343-3347.10.1016/0031-9422(91)83206-Z.
3) Clery, R. A., Cason, J. R.and Zelenay, V., (2016) Constituents of Cypriol Oil (Cyperus scariosus R.Br.): N-Containing Molecules and Key Aroma Components. J Agric Food Chem 64, 4566-73.10.1021/acs.jafc.6b00680.
4) WO2024023169 - CHROMEN-5-ONE DERIVATIVES FOR USE AS FRAGRANCES. ZHOU, Lijun; CHAI, An ;YU, Zhiyu. 2023

OR14J1 – the OR for patchoulol



Screening of 486 OR and OR variants containing the optimized C-terminal domain with 50 µM patchoulol



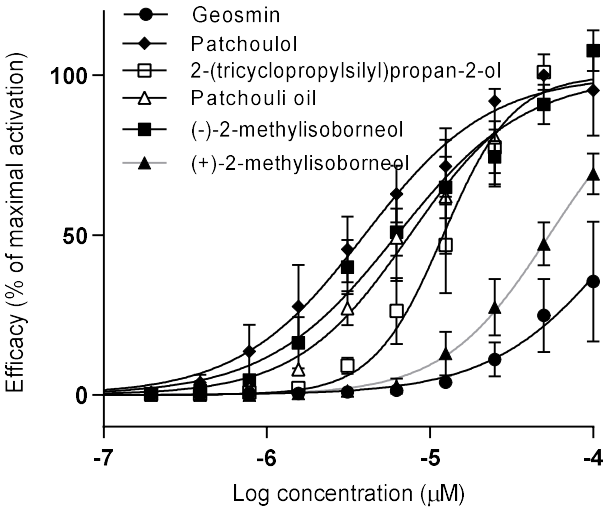
Activation and cell surface expression of OR14J1: Wild-type vs. optimized C-terminal sequence

Patchoulol is the key odorant in natural patchouli oil, with an odor described as 'woody, patchouli, earthy'. It strongly activated OR14J1 in the pan-OR screen, next to weaker activation of OR11A1 activated by earthy odors and the woody OR7A17. The wild-type of OR14J1 is inactive in HEK293 cells.

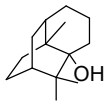
Among the library of 438 single fragrance ingredients, only natural patchouli oil activated this OR. This is in line with the fact that in order to obtain a patchouli smell, industry still has to rely on the natural oil, as no commercial synthetic analogues are available.

A rare example of a synthetic odorant with the distinct patchouli smell is 2-(tricyclopropylsilyl)-propan-2-ol [1]. This distinct structure has all the woody, patchouli and earthy aspects of natural patchoulol, with an around 5-10-fold lower potency *in vivo*. Indeed it activated OR14J1 with a 5-fold lower potency as compared to patchoulol, and it validates OR14J1 as the key Patchouli OR.

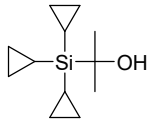
Interestingly, the earthy odorant (-)-2-methylisoborneol (MIB) also activates OR14J1. It has a patchouli odor to some individuals while for most the earthy, musty odor prevails, indicating it may bind yet to a further OR. Indeed. A recent patent application[4] reports activation of mammalian consensus OR8S1 by MIB. OR8S1 is polymorphic in humans, if human OR8S1 indeed is the additional OR respnding to MIB, the polymorphism may explain why it is perceived differently by different individuals.



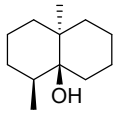
Activation of OR14J1 by patchouli and earthy odorants



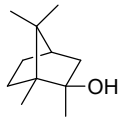
patchoulol
woody, patchouli, earthy
OTH: 0.09 ng/L
GTH: 0.02 / 0.07 ng
EC20% OR14J1: 1.2 x 10^{-6} M



2-(tricyclopropylsilyl)propan-2-ol
woody patchouly, resinous, earthy
OTH: 1.0 ng/L
GTH: 0.14 ng
EC20% OR14J1: 6.1 x 10^{-6} M



geosmin
earthy, musty
GTH: 0.001 ng
Ca. 0.02 ppb in water [2]
EC20% OR14J1: 4.6 x 10^{-5} M

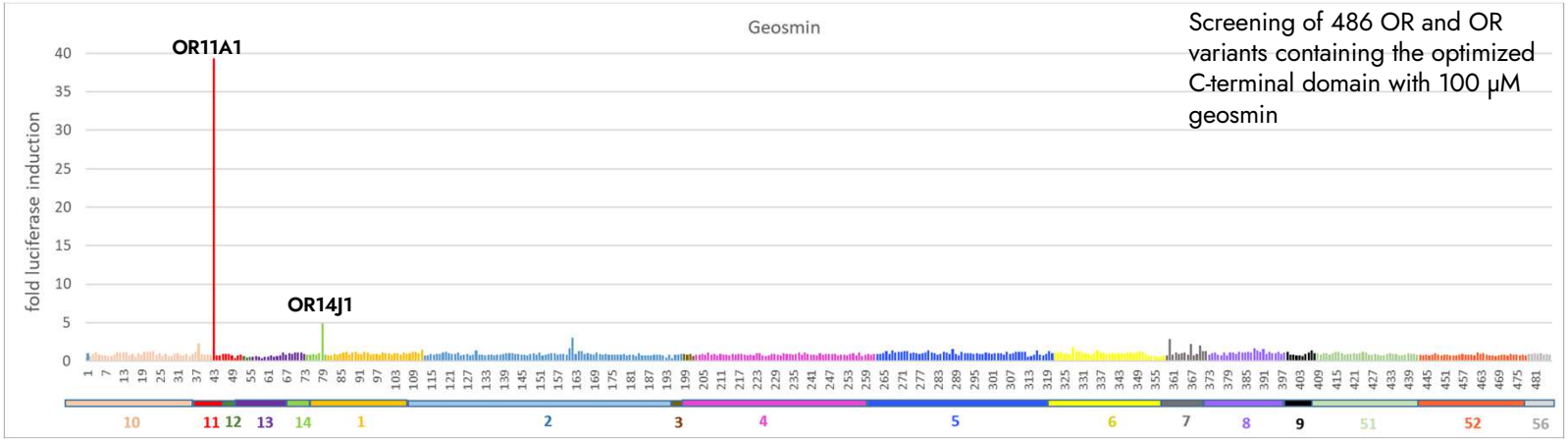


(±)-2-methyl isoborneol
earthy, musty, patchouli
GTH: 0.00016 ng
Ca. 0.002 - 0.02 ppb in water [3]
(-)-isomer: EC20% OR14J1: 1.8 x 10^{-6} M
(+)-isomer: EC20% OR14J1: 1.9 x 10^{-5} M

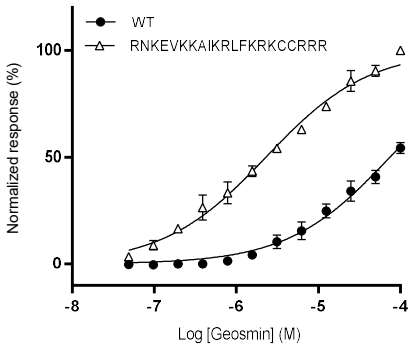
Structure, sensory data, and OR activation of ligands tested on OR14J1

1) Sunderkotter, A., S. Lorenzen, R. Tacke and P. Kraft (2010). "Novel silicon-based patchouli odorants of the trialkyl(1-hydroxy-1-methylethyl)silane type: design, synthesis, and olfactory properties." Chemistry **16**(25): 7404-7421. DOI: 10.1002/chem.201000549.
2) Polak, E. H.and Provasi, J., (1992) Odor sensitivity to geosmin enantiomers. Chem. Senses 17, 23-26.10.1093/chemse/17.1.23.
3) Lin, Tsair-Fuh; Watson, Susan; Dietrich, Andrea M.; (Mel) Suffet, I. H. (15 March 2018). Taste and Odour in Source and Drinking Water: Causes, Controls, and Consequences. 10.2166/9781780406664_0063
4) Patent application JP2024179802A by Kao corporation, ethod for searching for fragrance materials that suppress mold odor.

OR11A1 – the OR for earthy notes



Screening of 486 OR and OR variants containing the optimized C-terminal domain with 100 µM geosmin



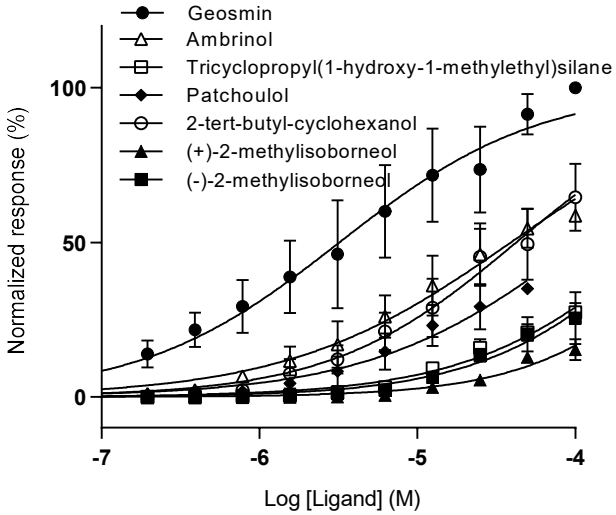
Activation or OR11A1: Wildtype sequence
Compared to the optimized C-terminal sequence

Geosmin is a highly potent odorant, giving the earthy note to beetroot and is often involved in water contaminated with off-odors[1].

Geosmin strongly and selectively activates OR11A1, next to weak activation of the patchouli OR14J1. OR11A1 is active in HEK293T cells with the WT sequence as observed before[2], but the optimized variant has 33-fold increased sensitivity.

Among 438 single fragrance ingredients, ambrinol and 2-tert-butyl cyclohexanol were identified as weak ligands. Interestingly, (-)-2-methyl isoborneol, another high-impact water contaminant with an 'earthy' odor description[3], only weakly activates this OR.

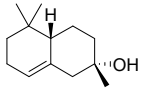
The patchouli odorants patchoulol and 2-(tricyclopropylsilyl)propan-2-ol both also have an earthy connotation and weakly activate this OR.



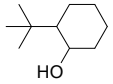
Activation of OR11A1 by earthy odorants



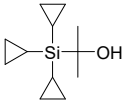
geosmin
earthy, musty
GTH: 0.001
OTH: Ca. 0.02 ppb in water [1]
EC20% OR11A1: 4.3 x 10^{-7} M



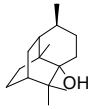
ambrinol
earthy, animalic, camphorated
OTH: 17.5 ng/L
GTH: 2 ng
EC20% OR11A1: 4 x 10^{-6} M



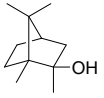
(±)-2-tert-butyl cyclohexanol
piny, camphoraceous, earthy-patchouli
OTH: 17.3 ng/L
EC20% OR11A1: 6.5 x 10^{-6} M



2-(tricyclopropylsilyl)propan-2-ol
woody patchouly, resinous, earthy
OTH: 1.04 ng/L
GTH: 0.14 ng
EC20% OR11A1: 5.1 x 10^{-5} M



patchoulol
woody, patchouli, earthy
OTH: 0.09 ng/L
GTH: 0.02 / 0.07 ng
EC20% OR11A1: 1.3 x 10^{-5} M

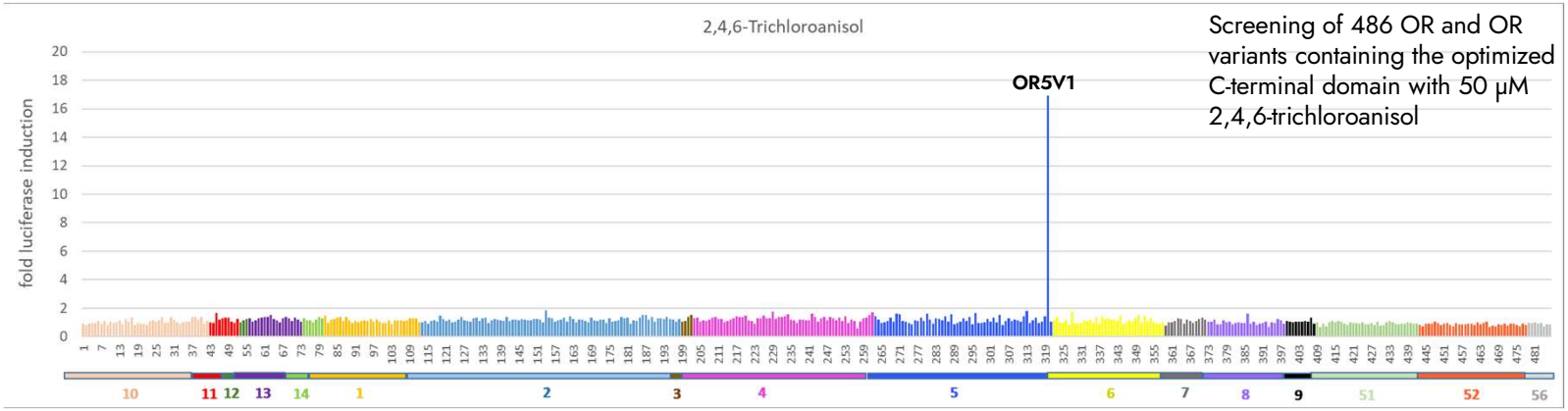


(±)-2-methyl isoborneol
earthy, musty
OTH: Ca. 0.002 - 0.02 ppb in water [3]
GTH: 0.00016
C20% OR11A1: 1.1 x 10^{-4} M

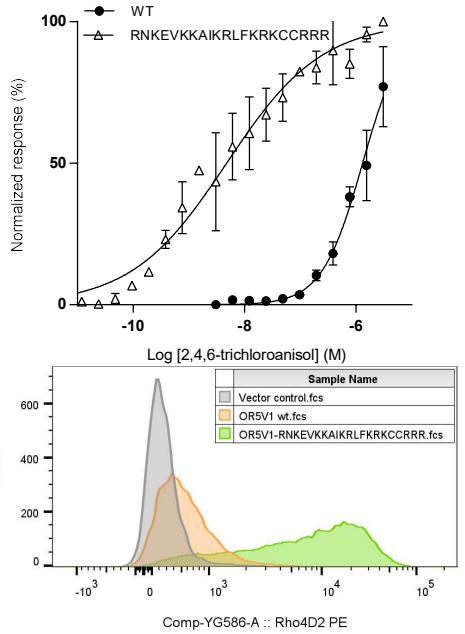
Structure, sensory data, and OR activation of earthy odorants tested on OR11A1

1) Polak, E. H.and Provasi, J., (1992) Odor sensitivity to geosmin enantiomers. Chem. Senses 17, 23-26.10.1093/chemse/17.1.23.
2) Ball, L. et al. 2024. Geosmin, a Food- and Water-Deteriorating Sesquiterpenoid and Ambivalent Semiochemical, Activates Evolutionary Conserved Receptor OR11A1. J. Agri. Food Chem. https://doi.org/10.1021/acs.jafc.4c015
3) Lin, Tsair-Fuh; Watson, Susan; Dietrich, Andrea M.; (Mel) Suffet, I. H. (15 March 2018). Taste and Odour in Source and Drinking Water: Causes, Controls, and Consequences. 10.2166/9781780406664_0063

OR5V1 – the OR for the cork note in tainted wine



Screening of 486 OR and OR variants containing the optimized C-terminal domain with 50 µM 2,4,6-trichloroanisol

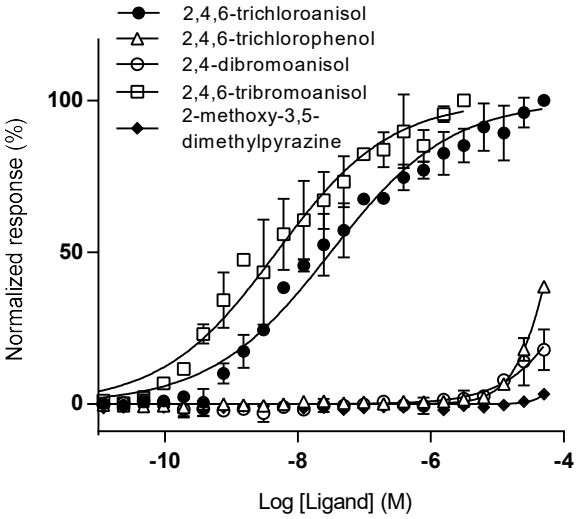


Activation and cell surface expression of OR5V1 Wild-type vs. optimized C-terminal sequence

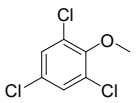
2,4,6-trichloroanisol (TCA) is a unique odorant responsible for most incidences of cork-tainted wines[1] but also occurs as important odoriferous contaminant in drinking water and packages. It has a uniquely low olfactive detection threshold[1,2]. A similar odor and even stronger odor impact has been reported for the tribromo analogue (TBA)[3]. In the pan-OR screening, TCA highly selectively activated only OR5V1. Activation starts at 10^{-9} M, while TBA was even more potent with activity in the high picomolar range. Removal of the methyl-group from TCA or of one bromine atom from TBA reduces potency by almost 4 orders of magnitude proving the high specificity of this OR. In addition, among 438 single fragrance ingredients, no additional ligand for OR5V1 was identified further underlining the unique specificity for this OR.

2-methoxy-3,5-dimethylpyrazine has also been reported as a cork-note in wine[4], yet with a distinct odor profile. Testing this compound on OR5V1 proves that yet another OR is involved in its detection.

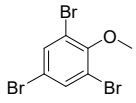
In principle OR5V1 could have been discovered with the wildtype sequence, but the optimized sequence has a 240-fold higher sensitivity.



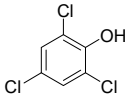
Activation of OR5V1 by different ligands



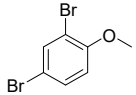
2,4,6,-trichloroanisol
OTH <10 ng/L in white wine [1)]
0.03-0.005 ng/L in water [2)]
EC20% OR5V1: 1.7 x 10^{-9} M



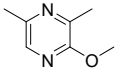
2,4,6,-tribromoanisol
OTH: 0.02 ng/L water [3)]
EC20% OR5V1: 3.1 x 10^{-10} M



2,4,6,-trichlorophenol
EC20% OR5V1: 2.7 x 10^{-5} M



2,4-dibromoanisol
EC20% OR5V1:5.3 x 10^{-5} M

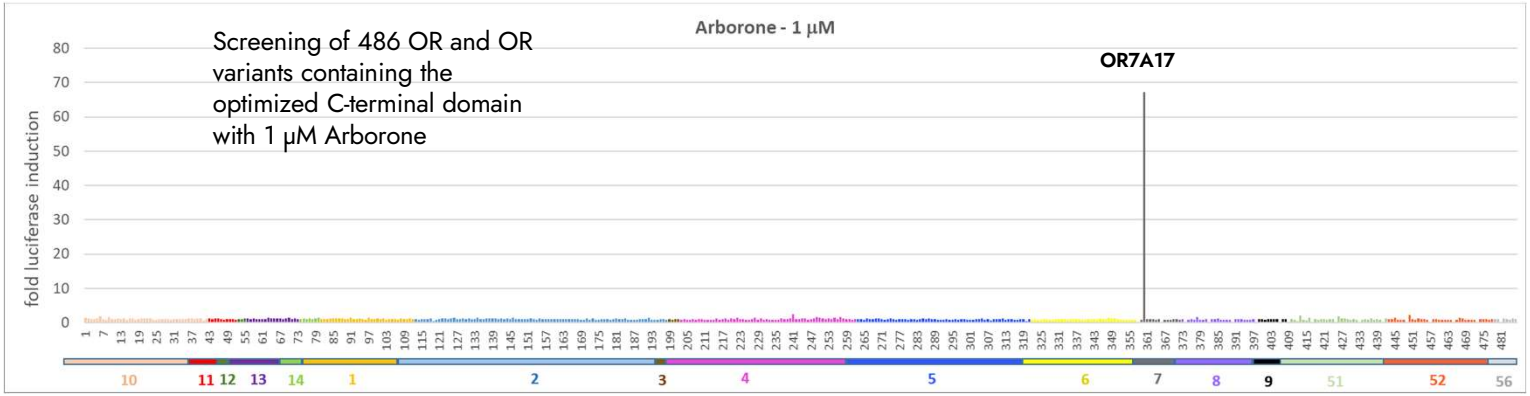


2-methoxy-3,5,dimethylpyrazine
OTH: 2.1 ng/L white wine [4)]
EC20% OR5V1: inactve

Structure, sensory data, and OR activation of ligands tested on OR5V1

1) Buser, H. R., Zanier, C.and Tanner, H., (1982) Identification of 2,4,6-Trichloroanisole as a Potent Compound Causing Cork Taint in Wine. J. Agric. Food Chem. 30, 359-362.10.1021/jf00110a037.
2) Barczak, R. J., Fisher, R. M., Le-Minh, N.and Stuetz, R. M., (2019) Importance of 2,4,6-Trichloroanisole (TCA) as an odorant in the emissions from anaerobically stabilized dewatered biosolids. Chemosphere 236.10.1016/j.chemosphere.2019.07.071.
3) Whitfield, F. B., Hill, J. L.and Shaw, K. J., (1997). 2,4,6-Tribromoanisole: A Potential Cause of Mustiness in Packaged Food. J. Agric. Food Chem. 45, 889-893. 10.1021/jf960587u
4) Chatonnet, P., Fleury, A.and Boutou, S., (2010) Origin and incidence of 2-methoxy-3,5-dimethylpyrazine, a compound with a "fungal" and "corky" aroma found in cork stoppers and oak chips in contact with wines. J Agric Food Chem 58, 12481-90.10.1021/jf102874f.

OR7A17 – the woody OR



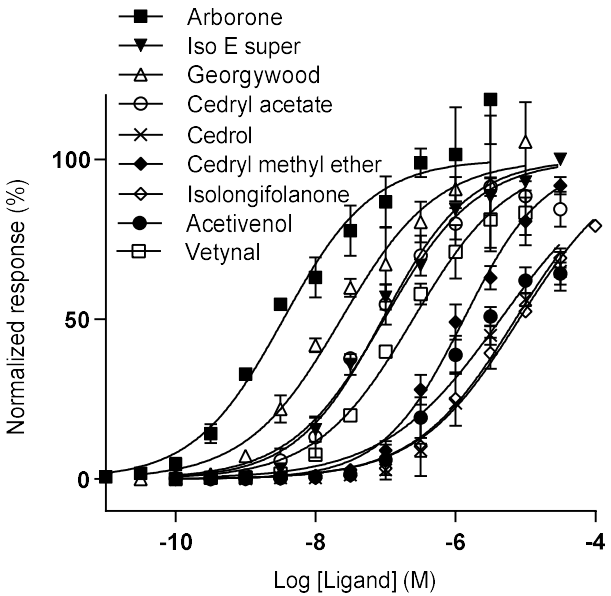
OR7A17 had been described as an OR activated by woody odorants in a patent application[1]. Arborone is known as the most potent woody odorant[2]. It selectively activates only OR7A17 among the 486 ORs in the library. The optimized C-terminus increases sensitivity of the detection in HEK293 cells > 400 fold.

Racemic Arborone is the most active odorant tested on OR7A17. It is contained at ca. 5% as the odor vector in the commercial product Iso E Super, which indeed is 30-fold less potent on OR7A17

The natural sesquiterpene cedrol is a major component of cedarwood oil with a woody odour. However, it has a high OTH which is paralleled by a high EC20% for OR7A17. Different derivatives of cedrol are commercially used as woody perfumery ingredients. Thus cedryl methyl ether has a lower EC20% and OTH. Cedryl acetate is a potent ligand for OR7A17, however, this is not fully reflected in the olfactive detection threshold.

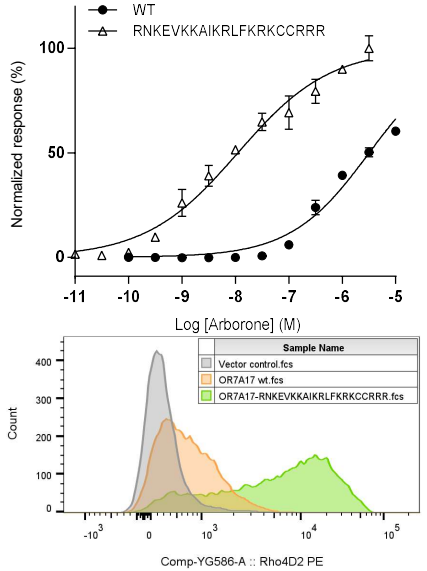
Among the library of 438 single fragrance ingredients, 20 ingredients with a 'woody' odor description activate OR7A17. This OR is quite tolerant in its ligand profile, and at high screening concentration (10^{-4} M) a variety of chemicals can weakly activate this OR. However, the potent activation in the low nanomolar range by the most potent woody odorant arborone and the SAR clearly validates it as a key OR for the perception of dry-cedarwood type woody odours.

OR7A17 activation was tested with both the optimized C-terminal variant and with the 1th generation optimisation RNKEVKDALKRLLKRKCC.

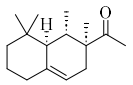


Activation of OR7A17 by woody odorants

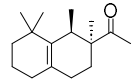
Data measured with the optimized variant. Note that both detetion thresholds in vivo and the in vitro activation span four orders of magnitude.



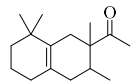
Activation and cell surface expression of OR7A17: Wild-type vs. optimized C-terminal sequence



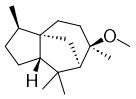
(±)-Arborone
woody, ambery, dry
GTH: 0.015 ng
EC20% OR7A17: 4.5×10^{-10} M
EC20% OR7A17*: 1.7×10^{-9} M



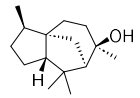
Georgywood (rac.) [1]
Cedarwood, woody-ambery
OTH: 0.14 ng/L
EC20% OR7A17: 2.6×10^{-9} M
EC20% OR7A17*: 8.2×10^{-9} M



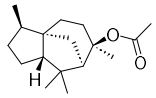
Iso E super (rac.) [1]
woody-ambery
OTH: 0.8 ng/L
GTH: 1.6 ng
EC20% OR7A17: 1.3×10^{-8} M
EC20% OR7A17*: 4.6×10^{-8} M



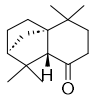
Cedryl methyl ether
Cedarwood, woody-ambery
OTH: 9.1 ng/L
EC20% OR7A17: 2.0×10^{-7} M
EC20% OR7A17*: 5×10^{-7} M



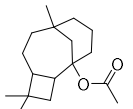
Cedrol
Cedarwood, woody
OTH: 78 ng/L
EC20% OR7A17: 7.3×10^{-7} M
EC20% OR7A17*: 2.2×10^{-6} M



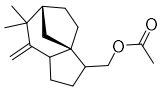
Cedryl acetate
Cedarwood, woody-ambery
OTH: 9.4 ng/L
GTH: 5.3 ng
EC20% OR7A17: 1.1×10^{-8} M
EC20% OR7A173*: 6.0×10^{-8} M



Isolongifolanone
woody
OTH: 163 ng/L
GTH: 72 ng
EC20% OR7A17: 7.9×10^{-7} M
EC20% OR7A17*: 1.4×10^{-6} M



Vetynal
sweet, dry, fresh, woody
GTH: 5.4 ng
OTH: 4.9 ng/L
EC20% OR7A17: 2.7×10^{-8} M
EC20% OR7A17*: 1.1×10^{-7} M



Acetivenol
woody
OTH: 6.1 ng/L
GTH: 9.2 ng
EC20% OR7A17: 3.0×10^{-7} M
EC20% OR7A17*: 5.6×10^{-7} M

Structure, sensory data, and OR activation of ligands tested on OR7A17

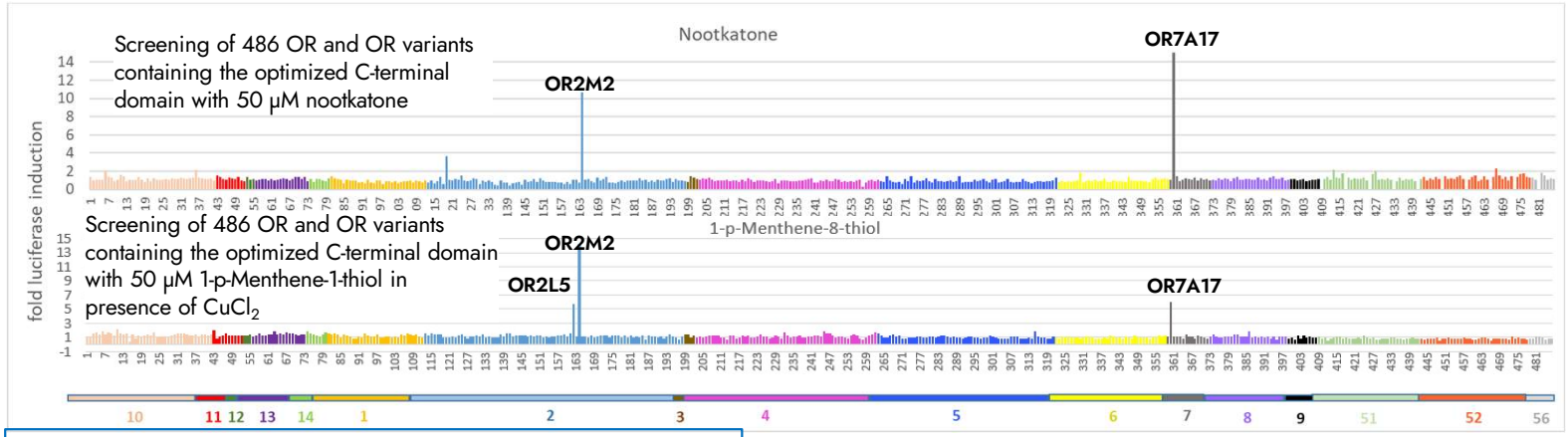
[1] These odorants are complex mixtures of isomers and only one key isomer is shown

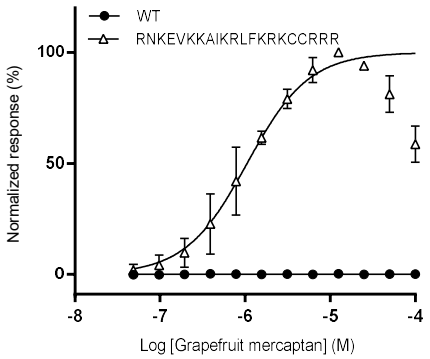
EC20% OR7A17*: measured with the 1th generation C-terminalvariant RNKEVKDALKRLLKRKCC

1) Method for attributing olfactory tonalities to olfactory receptor activation and methods for identifying compounds having the attributed tonalities. Patent application WO2021064201A1, https://patents.google.com/patent/WO2021064201A1/en

2) Hong, S. and E.J. Corey, Enantioselective syntheses of georgyone, arborone, and structural relatives. Relevance to the molecular-level understanding of olfaction. Journal of the American Chemical Society, 2006. 128(4): p. 1346-1352.

OR2M2 – the key OR for the detection of grapefruit notes





Activation of OR2M2: Wildtype sequence compared to the optimized c-terminal sequence

1-*p*-menthene-8-thiol ("grapefruit mercaptan", "thioterpineol") is an extremely potent constituent of grapefruit juice. It strongly activated OR2M2. This OR is inactive as WT and can only be expressed in HEK293 with the modified C-terminus leading to high surface expression. Weak activity was also noted for the fruity OR2L5 and the woody OR7A17.

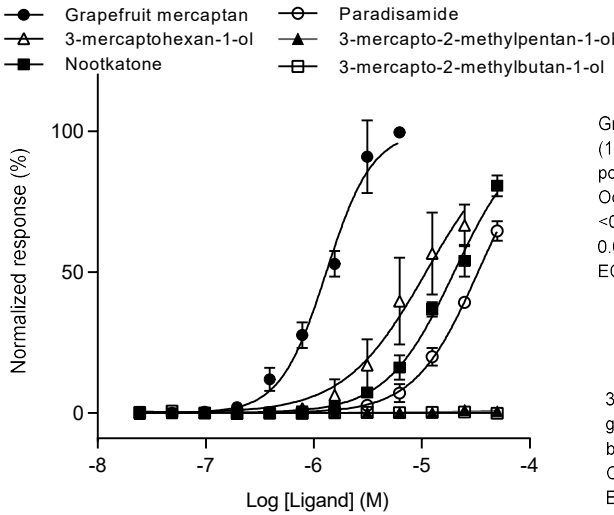
Similarly, the grapefruit constituent nootkatone selectively activates OR2M2 and to lesser extend OR7A17.

It is striking, that the structurally very different natural grapefruit odorants 1-*p*-menthene-8-thiol and nootkatone activate the same OR.

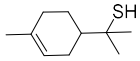
3-mercaptohexanol is widely described as giving the typical grapefruit note to e.g. Sauvignon blanc wines [4], and also strongly activates OR2M2.

Among the library of 438 single fragrance ingredients, nootkatone, decatone, paradisamide, and gardamide activated this OR. Despite the high structural diversity, all these odorants have the grapefruit odour note in common, indicating that OR2M2 is key for the odour nuance described as 'grapefruit' by human panelists.

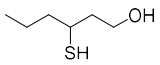
Decatone and nootkatone additionally activate OR7A17 explaining their woody aspect next to the grapefruit note.



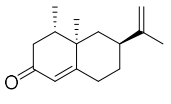
Activation of OR2M2 by grapefruit odorants



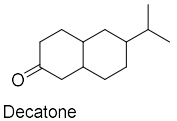
Grapefruit mercaptan
(1-p-menthene-8-thiol)
powerful, citrus, grapefruit and fruity
Odor threshold (Leffingwell):
<0.0001 ppb in water [1]
0.0003 ng/L in air [2]
EC20% OR2M2: 6.6 x 10^{-7} M [3]



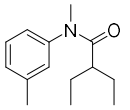
3-mercapto-hexanol
grapefruit note in Sauvignon blanc wine [4]
OTH: 0.08 ng/L [6]
EC20% OR2M2: 3.3 x 10^{-6} M



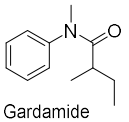
Nootkatone
fruity, citrus, woody
OTH: 2.1 ng/L
EC20% OR2M2: 7.4 x 10^{-6} M



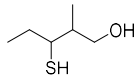
Decatone
Citrus, Grapefruit, Woody
Fruity, Fresh
OTH: 33 ng/L
EC20% OR2M2: 4.6 x 10^{-5} M



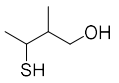
Paradisamide
fruity, fresh, grapefruit
OTH: 1.2 ng/L
GCT: 3.3 ng/L
EC20% OR2M2: 1.3 x 10^{-5} M



Gardamide
Citrus, Grapefruit, Rhubarb
OTH: 11 ng/L
GCT: 3 ng/L
EC20% OR2M2: 2.2 x 10^{-5} M



3-Mercapto-2-methylpentan-1-ol
Key food odorant in onions
OTH: 0.0014 ng/L [5]
EC20% OR2M2: inactive

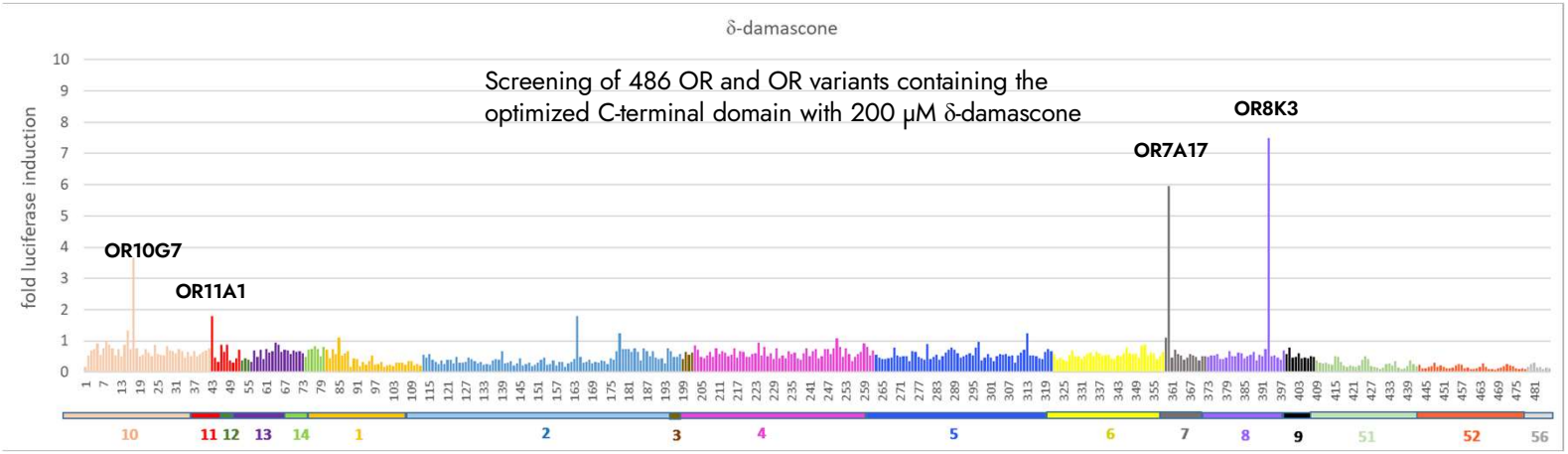


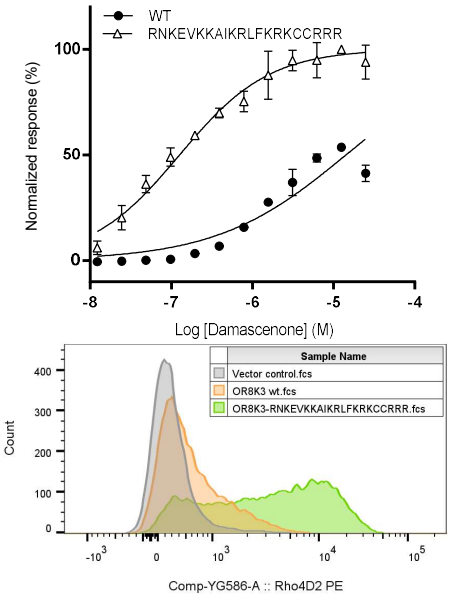
3-Mercapto-2-methylbutan-1-ol
Onion
OTH: 0.032 ng/L [5]
EC20% OR2M2: inactive

Structure, sensory data, and OR activation of ligands tested on OR2M2

1) Demole, E., Enggist, P.and Ohloff, G., (1982) 1-p-Menthene-8-thiol: A powerful flavor impact constituent of grapefruit juice (Citrus parodisi MACFAYDEN). *Helv. Chim. Acta 65,* 1785-1794.10.1002/hlca.19820650614
2) Schoenauer, S.and Schieberle, P., (2016) Structure-Odor Activity Studies on Monoterpenoid Mercaptans Synthesized by Changing the Structural Motifs of the Key Food Odorant 1-p-Menthene-8-thiol. J Agric Food Chem 64, 3849-61.10.1021/acs.jafc.6b01645.
3) Given are data based on nominal test concentration. 1-p-menthane-8-thiol is unstable in water and air and cyclises to (1S,4S,5S)-4,7,7-trimethyl-6-thiabicyclo[3.2.1]octane. Thus the actual EC20% and dose-response for the active constituent may still be below the given value.
4) Subileau, M., Schneider, R., Salmon, J. M.and Degryse, E., (2008) New insights on 3-mercaptohexanol (3MH) biogenesis in Sauvignon Blanc wines: Cys-3MH and (E)-hexen-2-al are not the major precursors. J Agric Food Chem 56, 9230-5.10.1021/jf801626f [doi]. 1
5) Noe, F. et al. OR2M3: A Highly Specific and Narrowly Tuned Human Odorant Receptor for the Sensitive Detection of Onion Key Food Odorant 3-Mercapto-2-methylpentan-1-ol. Chem Senses 42, 195-210, doi:10.1093/chemse/bjw118 (2017).
6) Steinhaus, M., Sinuco, D., Polster, J., Osorio, C. & Schieberle, P. Characterization of the aroma-active compounds in pink guava (Psidium guajava, L.) by application of the aroma extract dilution analysis. J Agric Food Chem 56, 4120-4127, doi:10.1021/jf8005245 (2008).

OR8K3 – the OR for damascenone





Activation and surface expression of OR8K3: Wildtype sequence compared to the optimized c-terminal sequence

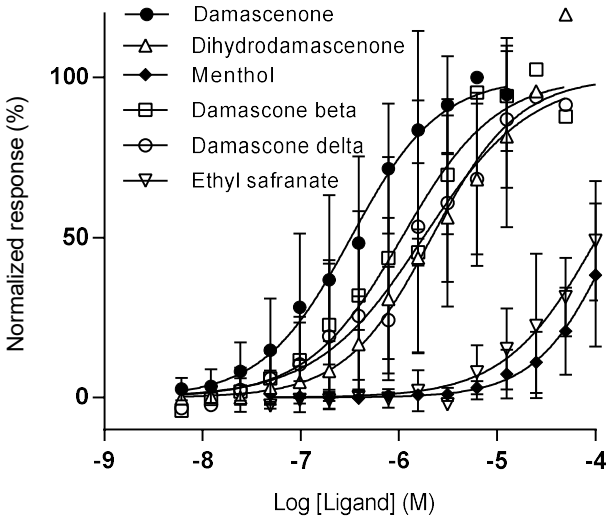
(E)-β-damascenone is a key high impact odorant in apple juice, with a typical cooked-apple smell[1] but it is equally an important odorant in red wines[2] and some essential oils. For the de-orphanisation screen, the commercial analogue δ-damascone was used. It activated most strongly OR8K3, and the DRP analysis indicated that indeed OR8K3 is a highly sensitive OR for (E)-β-damascenone and close analogues.

Weaker activation of the earthy OR OR11A1, the woody OR7A17 and the spicy OR10G7 was also noted at the high screening concentration of 200 µM. The wild-type of OR8K3 is almost two orders of magnitude less sensitive in HEK293 cells.

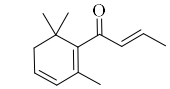
Among the library of 438 single fragrance ingredients, only damascone derivatives, the linear mimic (2E,5Z)-5,6,7-trimethylocta-2,5-dien-4-one and azabre activated this OR.

Dihydrodamascenone is a less potent agonist *in vitro* and *in vivo* – although the difference is higher *in vivo*.

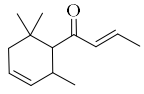
OR8K3 had initially be described as OR for menthol, but it is three orders of magnitude less sensitive towards menthol as compared to damascenone. Thus based on the studied analogues. OR8K3 appears to be the key OR for the typical scent of natural (E)-β-damascenone .



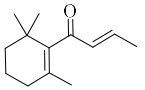
Activation of OR8K3 by different odorants



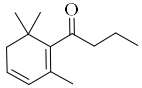
(E)–β-damascenone
dried fruits/winey
GTH: 0.0056 ng/L
EC20% OR8K3: 7.8 x 10^{-8} M



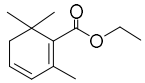
(E)-δ-damascone
dried fruits/winey
GTH: 0.0067 ng/L
EC20% OR8K3: 3 x 10^{-7} M



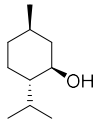
(E)-β-damascone
dried fruits/winey
GTH: 0.013 ng/L
EC20% OR8K3: 2.3 x 10^{-7} M



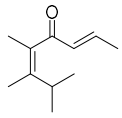
dihydro-β-damascenone
fruity, damascone, woody
GTH: 0.87 ng/L
EC20% OR8K3: 5.3 x 10^{-7} M



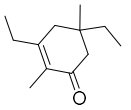
ethyl safranate
floral, rose, fruity, apple
OTH: 0.57 ng/L
EC20% OR8K3: 2.3 x 10^{-5} M



menthol
mint-like, fresh
OTH: 253 ng/L
EC20% OR8K3: 4.5 x 10^{-5} M



(2*E*,5*Z*)-5,6,7-trimethylocta-2,5-dien-4-one
floral, rosy, fruity, minty
OTH: 0.49 ng/L
EC20% OR8K3: 3 x 10^{-6} M



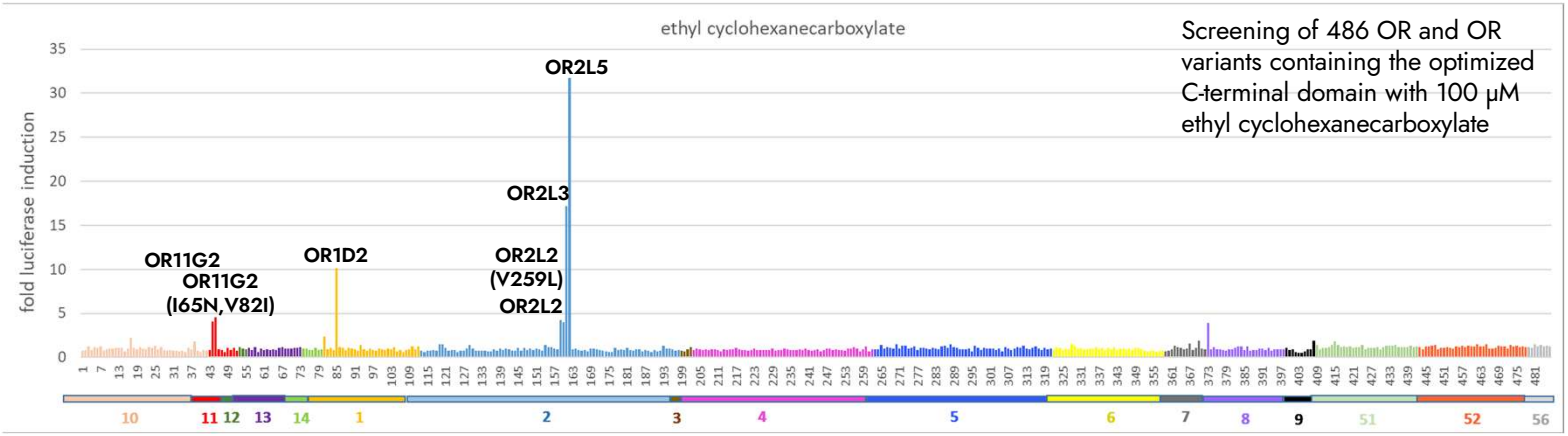
Azarbre -
21% activation at 10 µM
floral, woody, green
OTH: 11.56 ng/L

Structure, sensory data, and OR activation of ligands tested on OR8K3

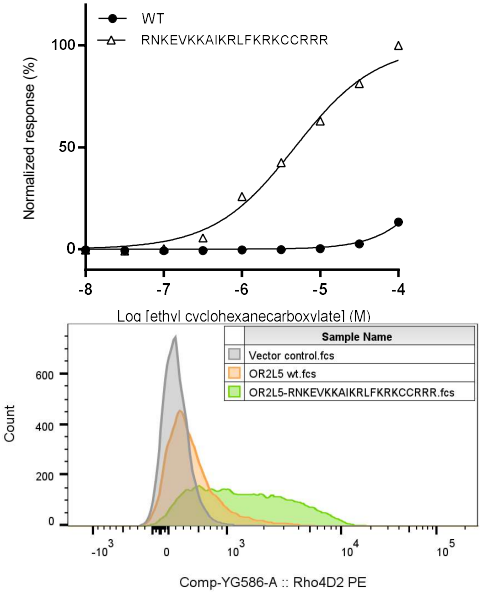
1) SENSORY EVALUATION OF COMMERCIAL APPLE JUICES AND RELATION TO SELECTED KEY AROMA COMPOUNDS. In:Expression of Multidisciplinary Flavour ScienceProceedings of the 12th Weurman Symposium. Edited by Imre Blank, Matthias Wüst, Chahan Yeretzian. Interlaken, Switzerland, 2008: p76-79.

2) Dunkel, A., Steinhaus, M., Kotthoff, M., Nowak, B., Krautwurst, D., Schieberle, P.and Hofmann, T., (2014) Nature's chemical signatures in human olfaction: a foodborne perspective for future biotechnology. Angew Chem Int Ed Engl 53, 7124-43.10.1002/anie.201309508.

OR2L5 – a highly selective OR for a high impact fruit ester



Screening of 486 OR and OR variants containing the optimized C-terminal domain with 100 µM ethyl cyclohexanecarboxylate

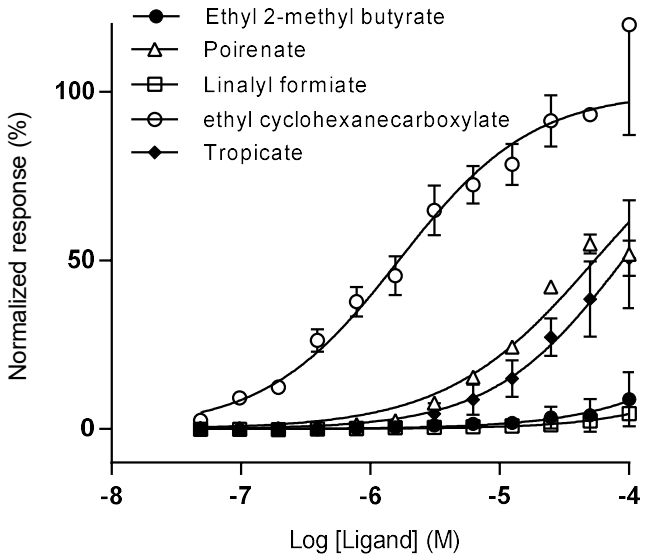


Activation and surface expression of OR2L5: Wildtype sequence compared to the optimized c-terminal sequence

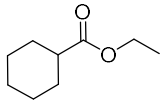
Ethyl cyclohexanecarboxylate is a one of the most potent fruity ester known, with an OTH of 0.003 ng/L and a GTH of 0.005 ng. It strongly activated OR2L5, with weaker activation for OR2L3. OR1D2 and OR11G2 are weakly activated, too, the latter is mainly responsive to ethyl 3-phenyl-2,3-epoxybutanoate, an ester with a strong strawberry note.

Among the library of 438 single fragrance ingredients, no additional ligand for OR2L5 was identified (except very weak activity for linalyl acetate only).

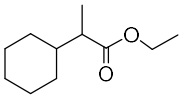
Two structurally closely related analogues (poirenate and tropicate) have an almost two orders of magnitude lower activity on OR2L5 along with a significantly higher OTH / GTH. It cannot be excluded that these also activate other OR, though.



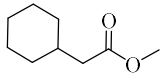
Activation of OR2L5 by small fruity esters



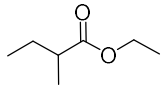
ethyl cyclohexanecarboxylate
fruity, overriped apple, red fruits
OTH: 0.003 ng/L
GTH: 0.005 ng
EC20% OR2L5: 3.2 x 10^{-7} M



Poirenate
fruity, green
OTH: 21 ng/L
GTH: 2.5 ng
EC20% OR2L5: 8.7 x 10^{-6} M



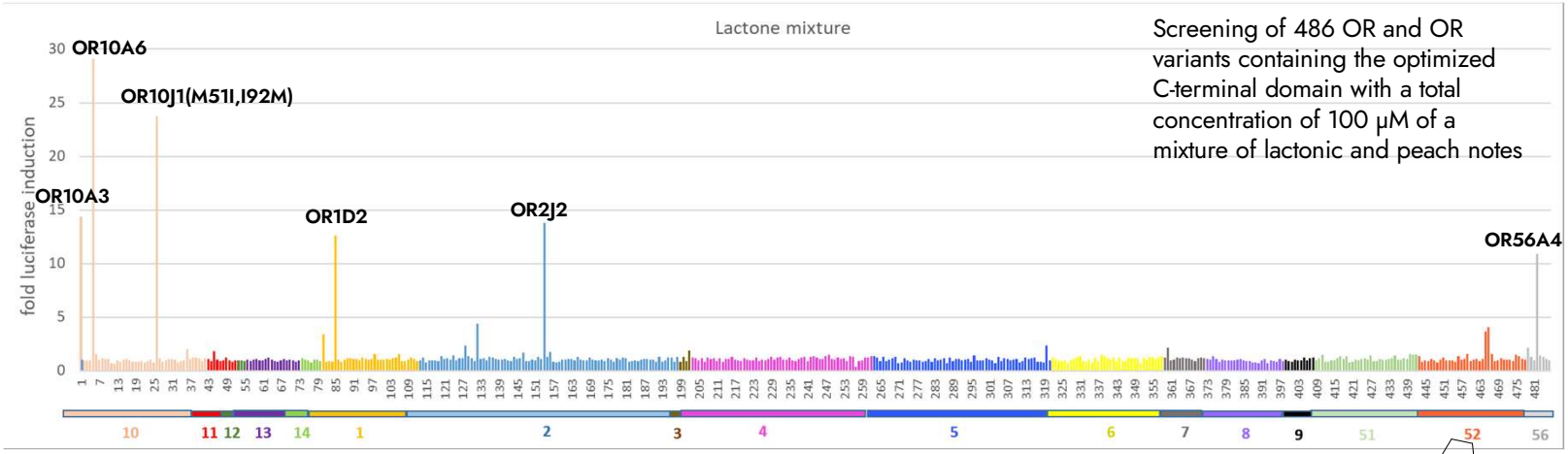
Tropicate
fruity, (tropical, pineapple)
GTH: 1.07 ng
EC20% OR2L5: 1.7 x 10^{-5} M



ethyl 2-methyl butyrate
green , fruity
OTH: 3.6 ng/L
EC20% OR2L5: inactive

Structure, sensory data, and OR activation of structurally related ligands tested on OR2L5

OR10A3 and OR10A6 – two key OR for lactonic and peach notes



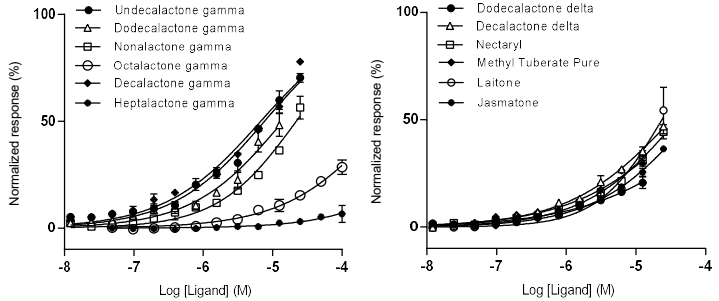
Screening of 486 OR and OR variants containing the optimized C-terminal domain with a total concentration of 100 µM of a mixture of lactonic and peach notes

Different lactones contribute to the odor of ripe peaches[1] with γ-decalactone contributing strongly to the peach aroma.

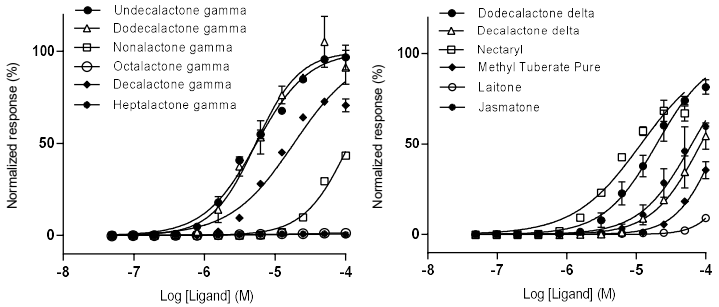
A mixture of lactones strongly activated OR10A3 and OR10A6, two closely related OR with 85% sequence identity. Additionally, some of these lactones activate OR10J1, as well as the more promiscuous OR1D2 and OR2J2. OR56A4 is activated by C8-C12 carboxylic acids, and most likely the open form of these lactones additionally activates this OR.

Based on the SAR and potency, OR10A3 and OR10A6 appear key OR to detect the longer chain lactonic notes at low concentration. γ-C10-C12 lactones, δ-dodecalactone and Nectaryl are the strongest ligands for OR10A6 and share the 'peach'-odor description, indicating that this OR appears particular important for the peach perception. An additional, not yet identified OR may be involved to detect the shorter lactones with a coconut odor note.

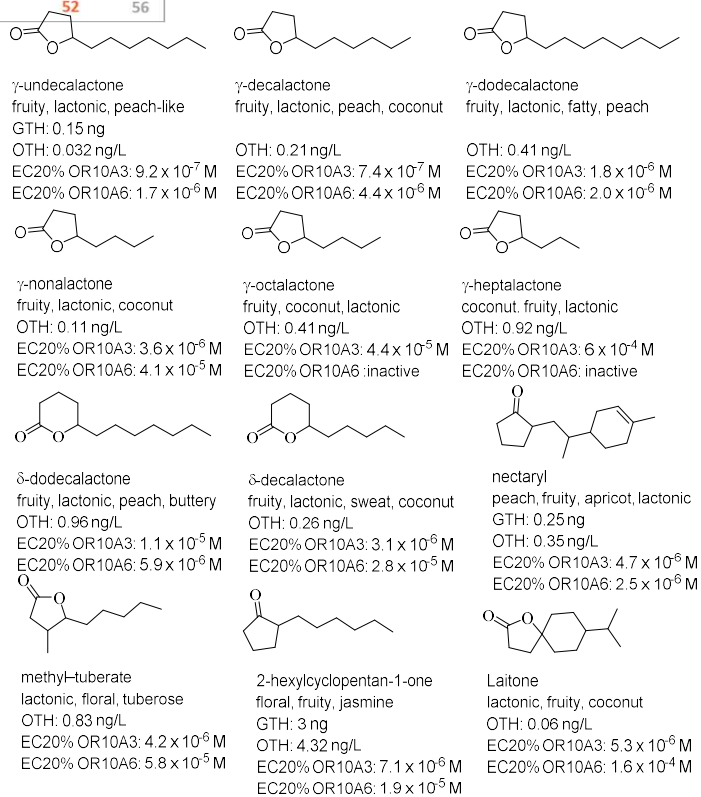
Among the library of 438 single fragrance ingredients, these lactones and two carba-analogues (2-heptylcyclopentan-1-one and 2-hexylcyclopentan-1-one) activated these OR, next to (E)-tridec-2-enenitrile. The latter has a quite distinct odor, and this hit is difficult to rationalize.



Activation of OR10A3 by lactonic odorants



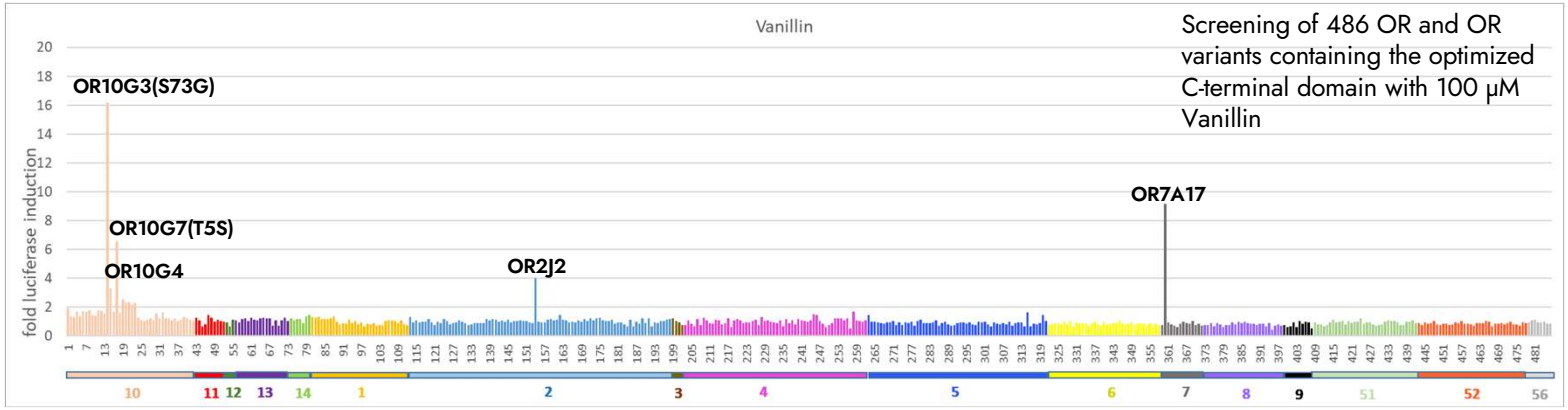
Activation of OR10A6 by lactonic odorants



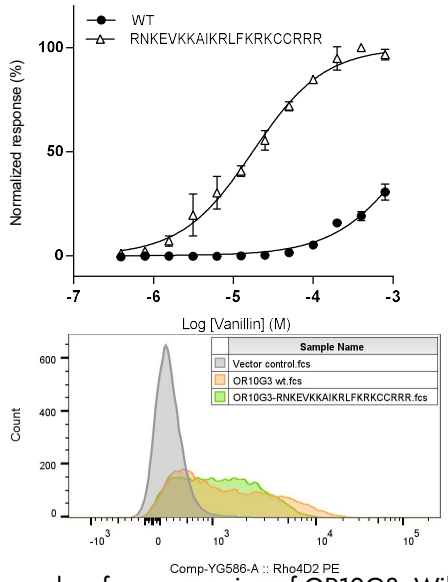
Structure, sensory data and OR activation of ligands tested on OR10A3 and OR10A6

1) Characterization of the Lactone Volatile Compounds in Different Types of Peach (Prunus persica L.) Fruit and Evaluations of Their Contributions to Fruit Overall Aroma. Scientia Agricultura Sinica ›› 2022, Vol. 55 ›› Issue (10): 2026-2037.doi: 10.3864/j.issn.0578-1752.2022.10.012

OR10G3 – the OR for vanilla notes



Screening of 486 OR and OR variants containing the optimized C-terminal domain with 100 µM Vanillin



Activation and surface expression of OR10G3: Wildtype sequence compared to the optimized c-terminal sequence

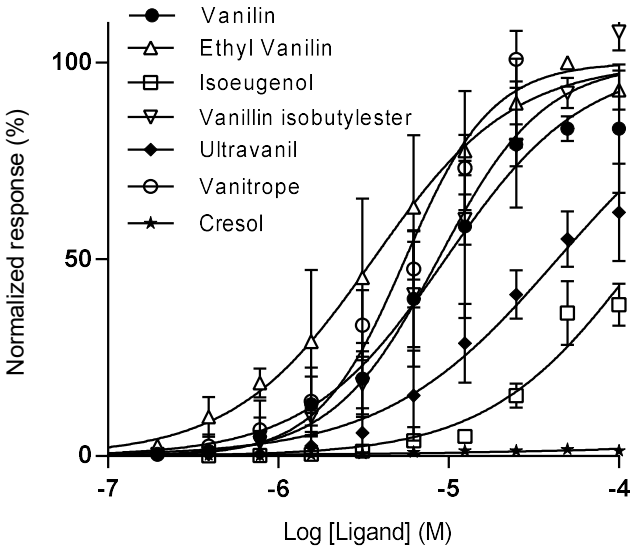
Vanillin is the signature odorant of vanilla beans. OR10G3 is the key OR activated by vanillin in the screening of the full OR-library.

Among the library of 438 single fragrance ingredients, additionally ethyl vanillin, vanillin isobutylester, ultravanil, vanitrope and isoeugenol activated this OR. All strong activators have the vanilla-smell in common. Ethyl vanillin is widely used commercially due to its more intense vanilla note, which is indeed reflected in a 2.5-fold lower EC20% on OR10G3 as compared to natural vanillin.

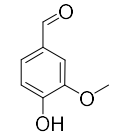
Interestingly, vanitrope is mainly activating OR10G3 and is inactive on OR10D3, despite close structural similarity to Isoeugenol. This is reflected by the sensory description as 'sweet, vanilla-like', which further underlines the importance of OR10G3 as OR for vanilla notes perceived in these structurally different odorants.

The wild-type OR10G3 is expressed in HEK293 cells and has been already implicated as one of several OR in vanillin detection [1)], however the optimized variant is two orders of magnitude more sensitive and the library screen and the SAR firmly places it as the key OR for the vanilla smell.

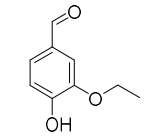
Previously, it was reported [1)] that vanillin equally activates OR10G4, however, OR10G4 is more specific to phenolic and smokey smelling analogues such as guaiacol (data not shown).



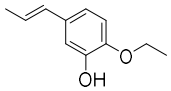
Activation of OR10G3 by vanilla- and phenolic odorants



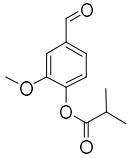
vanillin
balsamic, vanilla-like
GTH: 0.08 - 0.17 ng
OTH: 0.017 ng/L
EC20% OR10G3: 2.6 x 10^{-6} M



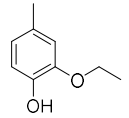
ethyl vanillin
balsamic, vanilla-like
GTH: 0.069 ng
OTH: 0.0097 ng/L
EC20% OR10G3: 9.8 x 10^{-7} M



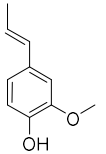
Vanitrope
balsamic, vanilla-like
OTH: 0.08 ng/L
EC20% OR10G3: 2.5 x 10^{-6} M



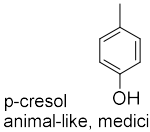
vanillin isobutylester
sweet, vanilla-like
GTH: 1 ng
OTH: 0.15 ng/L
EC20% OR10G3: 3.2 x 10^{-6} M [1)]



Ultravanil
vanilla-like, medicinal
GTH: 0.13 ng
OTH: 0.014 ng/L
EC20% OR10G3: 8.6 x 10^{-6} M [2)]



isoeugenol
Spicy, floral
GTH: 0.15 ng
OTH: 0.04 ng/L
EC20% OR10G3: 3.2 x 10^{-5} M [3)]



p-cresol
animal-like, medicinal
OTH: 0.8 ng/L
EC20% OR10G3: inactive

Structure, sensory data and OR activation of ligands tested on OR10G3

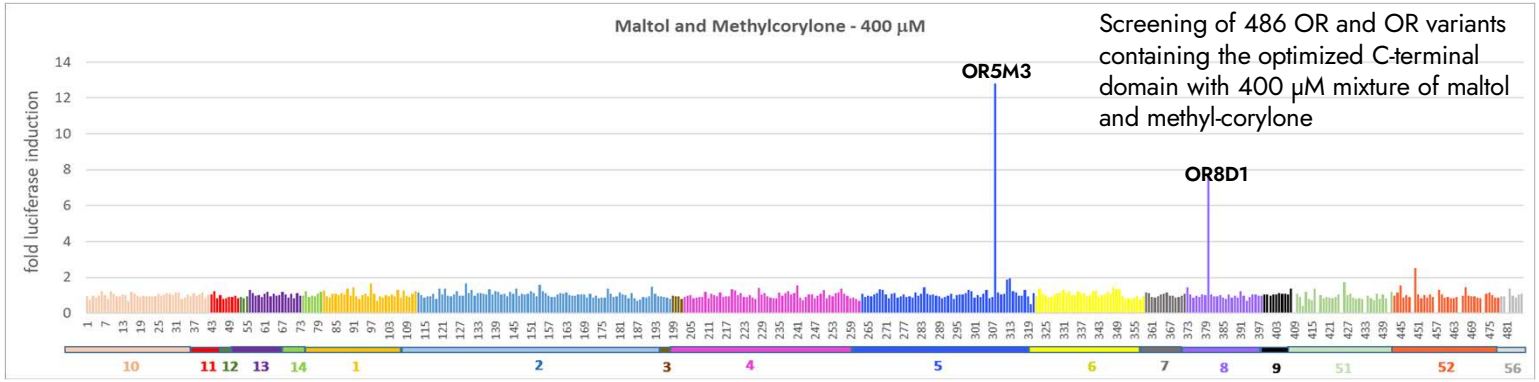
[1)] Veithen et al., 2017, Springer Handbook of Odor Chapter 22.2, Springer International publishing

[1)] Phenolic esters are rapidly cleaved in HEK cells (and potentially in the nose) – hence the signal may come from free vanillin

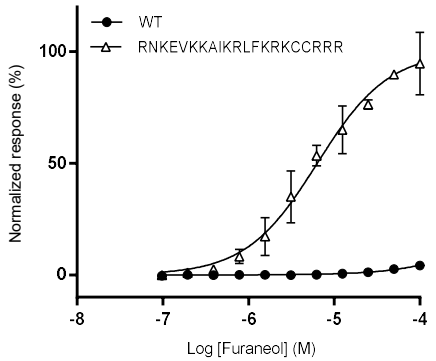
[2)] Ultravanil and Vanitrope are also powerfull activators of OR10G4, and less so of OR10G7 and 10G9, hence their detection threshold may be influenced by additional OR. On the other hand ethyl vanillin and vanillin are only weak activators of OR10G4, while OR10G3 is the main OR activated by these key vanilla notes.

[3)] Isoeugenol is mainly activating OR10G7 and OR10D3, see below

OR5M3 – the OR for sweet-caramelic notes and OR8D1 – the fenugreek OR



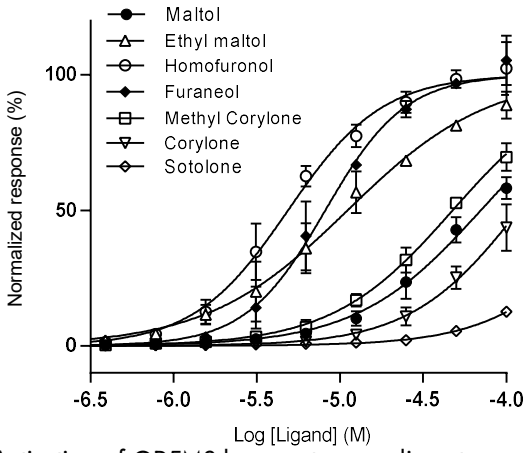
Screening of 486 OR and OR variants containing the optimized C-terminal domain with 400 µM mixture of maltol and methyl-corylone



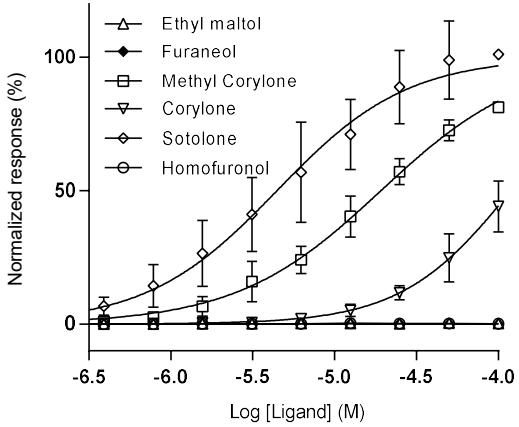
Activation or OR5M3: Wildtype sequence compared to the optimized C-terminal sequence

A mixture of the sweet caramelic notes maltol and methyl-corylone specifically activated OR5M3 and OR8D1. These OR had been found before to be activated by food furanones [1]). The current data indicate a broader spectrum of ligands with typical sweet-caramel notes like maltol also activating OR5M3, but with the exception of these sweet notes, no further ligands were identified among 438 commercial fragrance molecules. Furthermore, contrary to previous data on a more limited set of ligands, there is a continuum between these two OR with e.g. methyl-corylone having the properties to activate both OR.

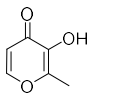
Also, the optimized C-terminal sequence now reveals the high sensitivity of OR5M3, with activation at an over two orders of magnitude lower concentration as compared to the wild-type.



Activation of OR5M3 by sweet-caramelic notes



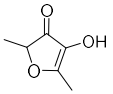
Activation of OR8D1 by sweet-caramelic notes



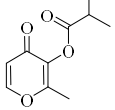
maltol
Sweet, caramel, balsamic, vanilla
OTH: 0.98 ng/L
EC20% OR5M3: 2.1×10^{-5} M
EC20% OR8D1: inactive



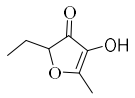
ethyl maltol
Sweet, caramel, balsamic
OTH: 0.42 ng/L
EC20% OR5M3: 2.9×10^{-6} M
EC20% OR8D1: inactive



furaneol
Sweet, caramel, fruity
OTH: 0.018 ng/L
EC20% OR5M3: 3.8×10^{-6} M
EC20% OR8D1: inactive



maltyl isobutyrate
Sweet, caramel, fruity, strawberry
OTH: 4.02 ng/L
EC20% OR5M3: 4.5×10^{-5} M
EC20% OR8D1: inactive



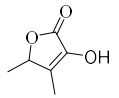
Homofuronol
Sweet, caramel, balsamic, fruity
OTH: 0.0.06 ng/L
EC20% OR5M3: 1.9×10^{-6} M
EC20% OR8D1: inactive



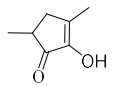
Methyl Corylone
burnt sugar-like, roasty, spicy, nutty
OTH: 0.0065 ng/L
EC20% OR5M3: 1.5×10^{-5} M
EC20% OR8D1: 4.7×10^{-6} M



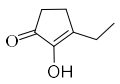
Corylone
burnt sugar, nutty, maple-like
OTH: 0.73 ng/L
EC20% OR5M3: 4.2×10^{-5} M
EC20% OR8D1: 4.1×10^{-5} M



Caramel lactone, Sotolone
Caramel, nut
OTH: 0.003 ng/L
EC20% OR5M3: 1.5×10^{-4} M
EC20% OR8D1: 1.2×10^{-6} M



Coronol (5-methyl-corylone)
Caramel, nut
OTH: 1.99 ng/L
EC20% OR5M3: 1.5×10^{-5} M
EC20% OR8D1: 1.2×10^{-4} M

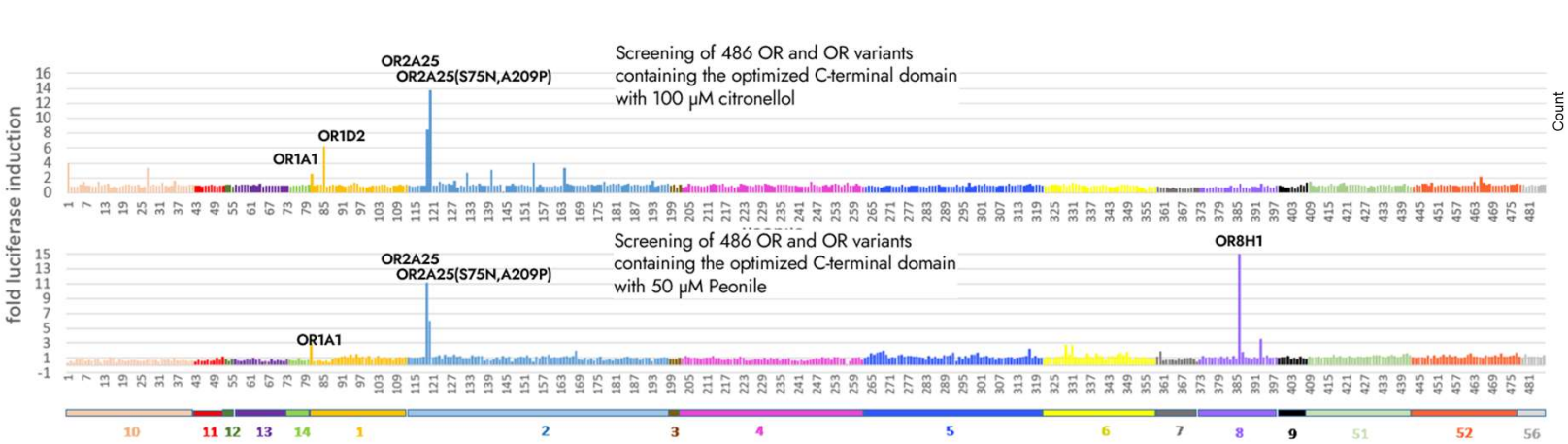


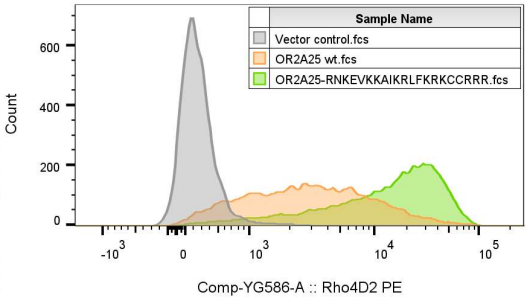
Ethylcyclotene
Caramel, nut
OTH/GTH: no data
EC20% OR5M3: 7.8×10^{-6} M
EC20% OR8D1: 8.7×10^{-5} M

Structure, sensory data and OR activation of ligands tested on OR5M3 and OR8D1

1) Haag, F., S. Hoffmann, and D. Krautwurst, Key Food Furanones Furaneol and Sotolone Specifically Activate Distinct Odorant Receptors. J Agric Food Chem, 2021. 69(37): p. 10999-11005.

OR2A25 – the common OR for rosy odorants



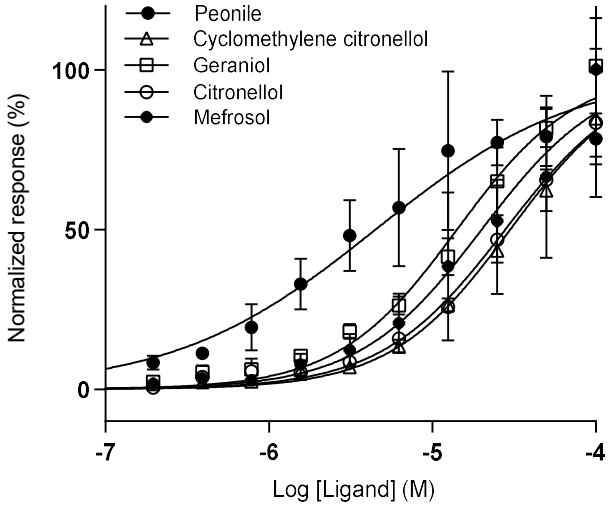


Surface expression of OR2A25: Wildtype sequence compared to the optimized C-terminal sequence

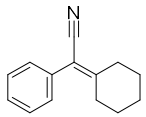
Geraniol and Citronellol are key natural signature odorant of rose petals and geranium oil. A rosy odor is also evoked by the structurally divergent synthetic odorant Peonile. The OR library was screened with Peonile and citronellol. OR2A25 is commonly activated by these two odorants. A series of related, natural and synthetic rose odorants including the natural signature rose odorant geraniol indeed activate this OR, too.

Peonile additionally activates OR8H1, which is the key OR activated by cyclohexyl salicylate (data not shown), and it therefore combines both the rosy aspects and the floral-green aspects of cyclohexyl salicylate.

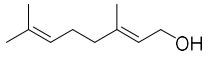
Thus, OR2A25 appears a common denominator to the key rosy aspect of these distinct chemical structures.



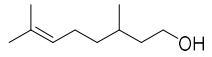
Activation of OR2A25 by rose odorants



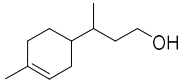
Peonile
Floral, Geranium
GTH: 0.4 ng
OTH: 0.6 ng/L
EC20% OR2A25: 6.3×10^{-7} M



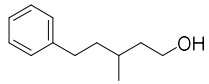
geraniol
Floral, rosy
GTH: ca. 1 ng
OTH: 2.7 ng/L
EC20% OR2A25: 4.4×10^{-6} M



citronellol
Floral, rosy
GTH: 3 ng
OTH: 3.4 ng/L
EC20% OR2A25: 8.3×10^{-6} M



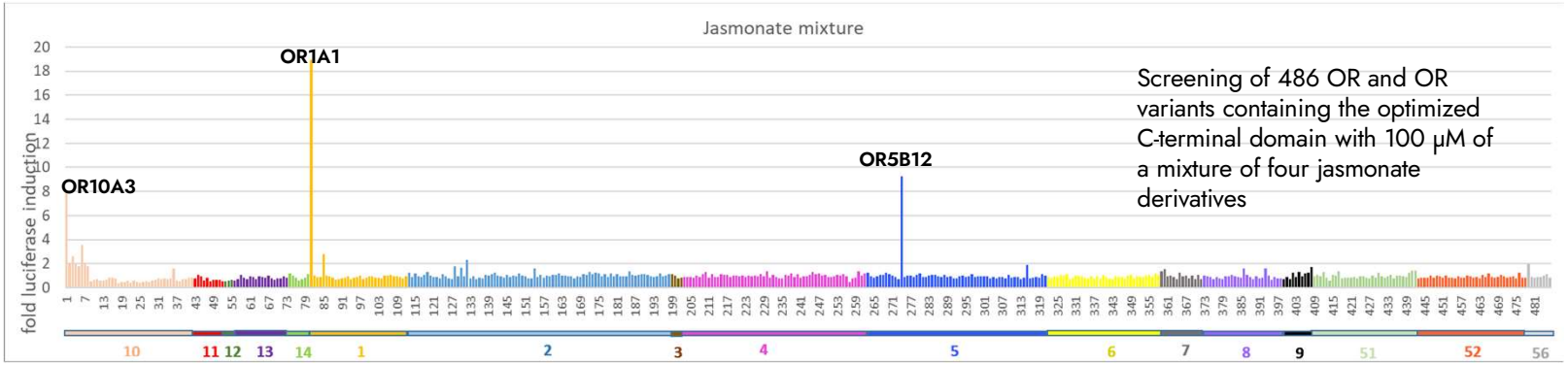
Cyclomethylene citronellol
Floral, mild
OTH: 17 ng/L
EC20% OR2A25: 9.4×10^{-6} M



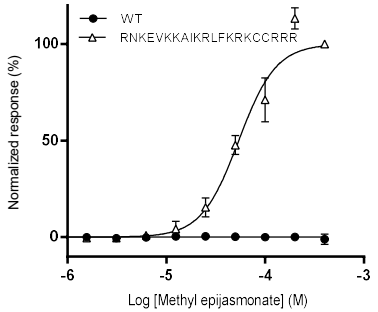
mefrosol
Floral, rosy
GTH: 0.87 ng/L
OTH: 0.77 ng/L
EC20% OR2A25: 6×10^{-6} M

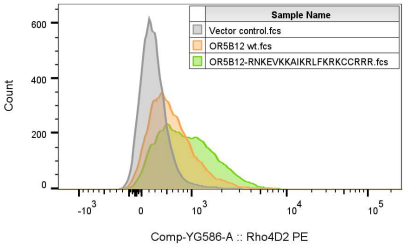
Structure, sensory data and OR activation of ligands tested on OR2A25

OR5B12 and OR1A1 – the OR for jasmonate derivatives



Screening of 486 OR and OR variants containing the optimized C-terminal domain with 100 µM of a mixture of four jasmonate derivatives



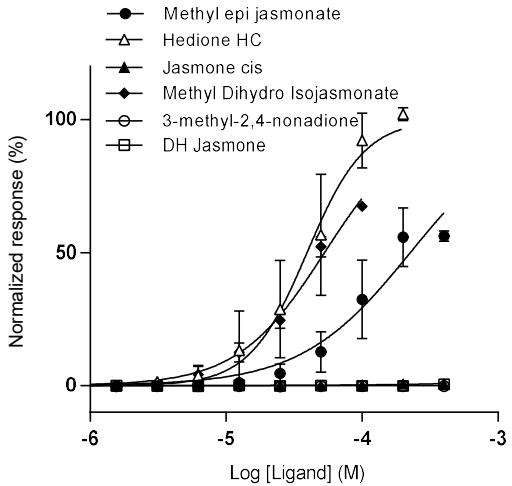


Activation and surface expression of OR5B12: Wildtype sequence compared to the optimized c-terminal sequence

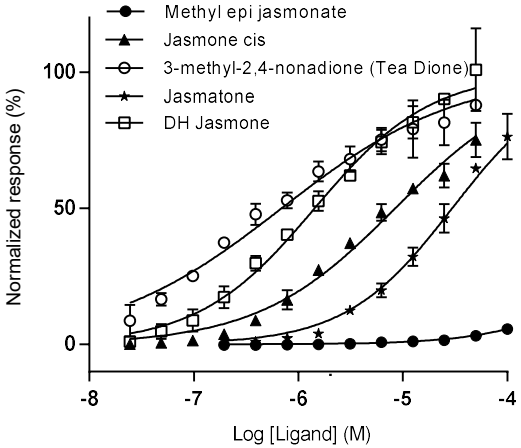
Methyl epi jasmonate, a derivative of the plant stress hormone jasmonic acid, is an important component of flower scents including jasmine and honeysuckle, and a key contributor to black tea and citrus peel aromas [1-3]. Its dihydro analogue is widely used commercially. *Cis*-jasmone is a product of jasmonic acid after oxidative decarboxylation[4] and is co-occuring in tea and jasmine with dihydrojasmone. A mixture of these compounds activated OR1A1 and OR5B12. Methyl epi jasmonate and its close analogues only activated OR5B12, while de-carboxylated analogues are selective on OR1A1.

Among 438 single commercial fragrance ingredients, only methyl dihydro jasmonate activated OR5B12, indicating a high specificity of this OR towards natural methyl epi jasmonate and its synthetic analogues.

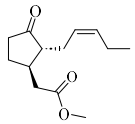
OR1A1 had initially been described as a broadly tuned OR, but later shown to be activated more specifically by the tea odorant 3-methyl-2,4-nonadione[4]. We confirmed this finding and show that this selectivity extends to jasmone derivatives also found in tea, and the sensitivity is 100-fold improved with the optimized C-terminus



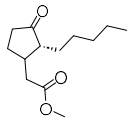
Activation of OR5B12 by jasmonate derivatives



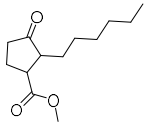
Activation of OR1A1 by jasmonate derivatives



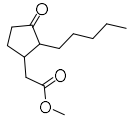
methyl epi jasmonate [a)]
floral, herbaceous, fresh
OTH: 0.45 ng/L
EC20% OR5B12: 6.3×10^{-5} M
EC20% OR1A1: $> 5 \times 10^{-4}$ M



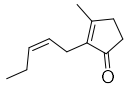
methyl dihydro jasmonate (high cis)[a)]
floral, citrus, fresh
OTH: 0.03 ng/L
EC20% OR5B12: 2.0×10^{-5} M
EC20% OR1A1: $> 5 \times 10^{-4}$ M



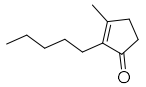
methyl dihydro isojasmonate[a)]
floral, jasmine-like, fresh
OTH: 0.12 ng/L
EC20% OR5B12: 1.9×10^{-5} M
EC20% OR1A1: $> 5 \times 10^{-4}$ M



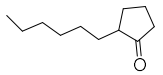
methyl dihydro jasmonate (racemic)
floral, jasmine-like, fresh
OTH: 0.1 ng/L
EC20% OR5B12: 1.7×10^{-5} M
EC20% OR1A1: $> 5 \times 10^{-4}$ M



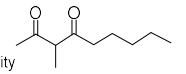
cis-jasmone
floral, jasmine-like
OTH: 2.9 ng/L
EC20% OR5B12: inactive
EC20% OR1A1: 1.0×10^{-6} M



dihydro jasmone[a)]
floral, jasmine-like
OTH: 0.62 ng/L
EC20% OR5B12: inactive
EC20% OR1A1: 2.4×10^{-7} M



Jasmatone
cis-Jasmone-like but more fruity
OTH: 4.3 ng/L
EC20% OR5B12: inactive
EC20% OR1A1: 5.9×10^{-6} M



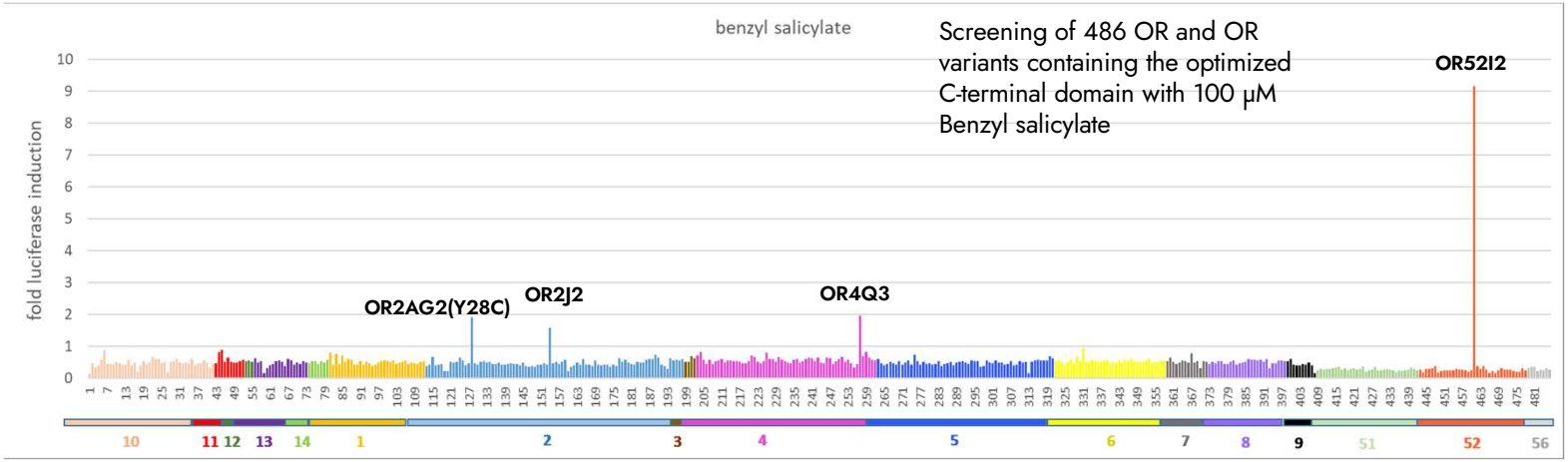
3-methyl-2,4-nonadione
green tea, fatty
OTH: 0.06 ng/L
EC20% OR5B12: inactive
EC20% OR1A1: 4.6×10^{-8} M

Structure, sensory data and OR activation of ligands tested on OR5B12 and OR1A1

a) These four ingredients were used as mixture to screen the OR library

1) Li, J., Hao, C., Jia, H., *et al.* (2022) Aroma characterization and their changes during the processing of black teas from the cultivar, Camellia sinensis (L.) O. Kuntze cv. Jinmudan. Journal of Food Composition and Analysis 108.10.1016/j.jfca.2022.104449.
2) Feng, Z., Li, M., Li, Y., Wan, X.and Yang, X., (2020) Characterization of the orchid-like aroma contributors in selected premium tea leaves. Food Res. Int. 129.10.1016/j.foodres.2019.108841.
3) Ikeda, N., Ishihara, M., Tsuneya, T., Kawakita, M., Yoshihara, M., Suzuki, Y., Komaki, R.and Inui, M., (1994) Volatile components of honeysuckle (lonicera japonica thunb.) Flowers. Flavour Fragrance J. 9, 325-331.10.1002/ffj.2730090609
4) Koch, T., Bandemer, K.and Boland, W., (1997) Biosynthesis of cis-Jasmone: A Pathway for the Inactivation and the Disposal of the Plant Stress Hormone Jasmonic Acid to the Gas Phase? Helv. Chim. Acta 80, 838-850.10.1002/hlca.19970800318..
5) Geithe, C., et al., The Broadly Tuned Odorant Receptor OR1A1 is Highly Selective for 3-Methyl-2,4-nonanedione, a Key Food Odorant in Aged Wines, Tea, and Other Foods. Chem Senses, 2017. 42(3): p. 181-193

OR52I2 – a selective OR for salicylate esters

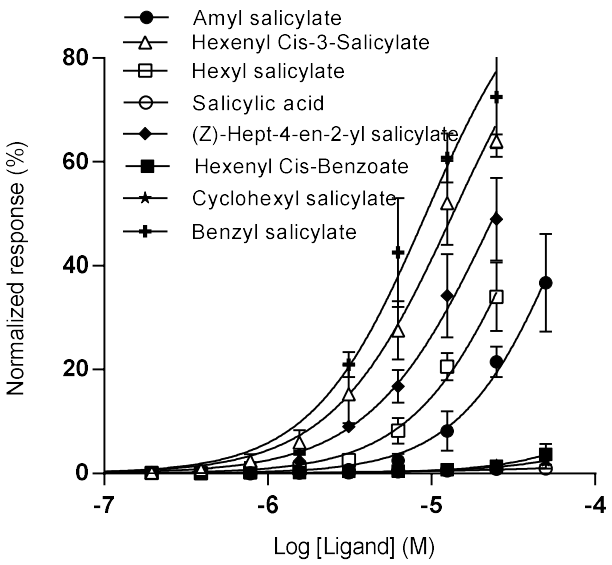


Salicylate esters, especially benzyl salicylate, naturally occur as floral odorants in white flowers, especially Ylang-Ylang but also tuberose and jasmine. Next to benzyl salicylate, a number of related esters occur naturally and are widely used in perfumery. Benzyl salicylate strongly activates OR52I2. This class I OR is very selective and only activated by salicylates among the 438 screened fragrant materials.

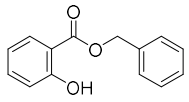
The potency for OR52I2 activation by these salicylates is strongly correlating to their OTH, with benzyl salicylate being most potent and amyl salicylate least potent both *in vivo* and for OR52I2 activation, validating this OR as the key OR to detect these salicylates.

Cyclohexyl salicylate has a different odour characteristic and indeed does hardly activate OR52I2, but it activates predominantly OR8K1 (data not shown). The wintergreen odorant methyl salicylate has a distinct odour, too, and does not activate OR52I2.

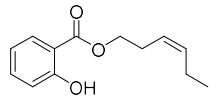
In general, class I OR are predominantly activated by carboxylic acids, and OR52I2 was therefore screened on a library of 42 odorant carboxylic acids. None of those activated OR52I2, and thus this OR is maybe the first reported exemption of a class I OR specifically activated by another structural class of odorants and not by carboxylic acids.



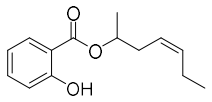
Activation of OR52I2 by salicylate esters



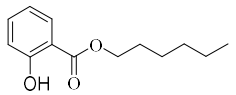
benzyl salicylate
floral, jasmine-like, balsamic
GTH: 0.38 ng
OTH: 0.15 ng/L
EC20% OR52I2: 3.04 x 10^{-6} M



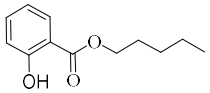
(Z)-hex-3-enyl salicylate
floral, jasmine-like, white-flower
OTH: 0.28 ng/L
EC20% OR52I2: 4.2 x 10^{-6} M



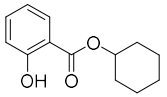
(Z)-hept-4-en-2-yl salicylate
floral, jasmine-like, white-flower
GTH: 0.77 ng
OTH: 0.35 ng/L
EC20% OR52I2: 7.3 x 10^{-6} M



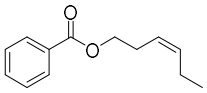
hexyl salicylate
floral, balsamic
OTH: 2.8 ng/L
EC20% OR52I2: 1.4 x 10^{-5} M



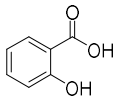
amyl salicylate
floral, balsamic
GTH: 4 ng
OTH: 5.2 ng/L
EC20% OR52I2: 2.6 x 10^{-5} M



cyclohexyl salicylate
floral, herbaceous
OTH: 4.5 ng/L
EC20% OR52I2: inactive



(Z)-hex-3-enyl benzoate
green, herbaceous
OTH: 150 ng/L
EC20% OR52I2: inactive



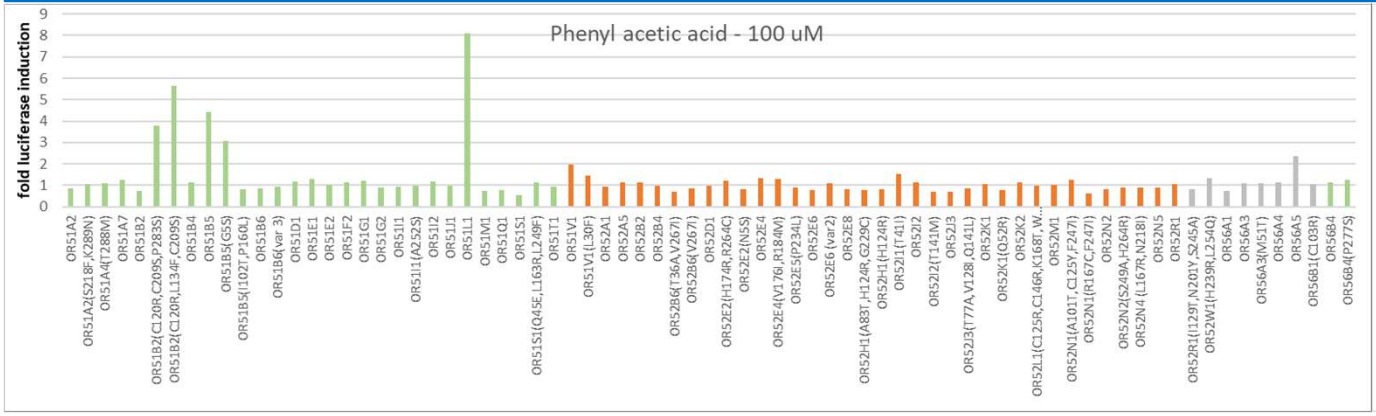
Salicylic acid
Odorless
EC20% OR52I2: inactive

Structure, sensory data and OR activation of ligands tested on OR52I2

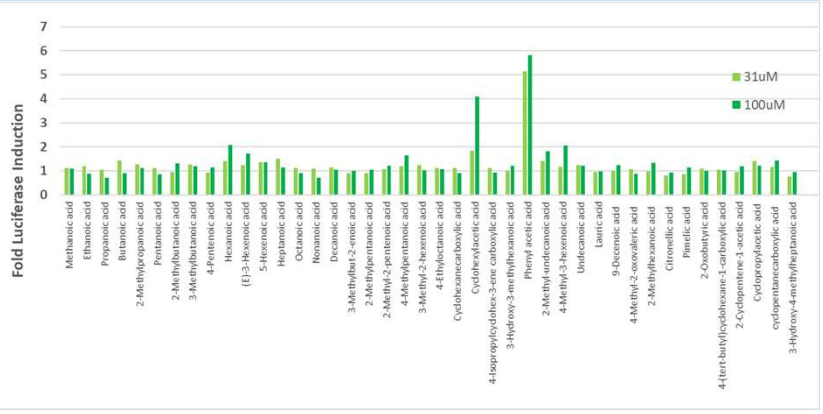
OR51L1 – the OR for phenylacetic acid

Phenylacetic acid is an odorant which has a sweet, honey and floral smell upon dilution and it naturally occurs in honey and many essential oils and flowers. Among class I OR, strongest activation was observed for OR51L1, and weaker activation for OR51B2 and OR51B5. No activation of any class II OR by phenylacetic acid was observed (data not shown).

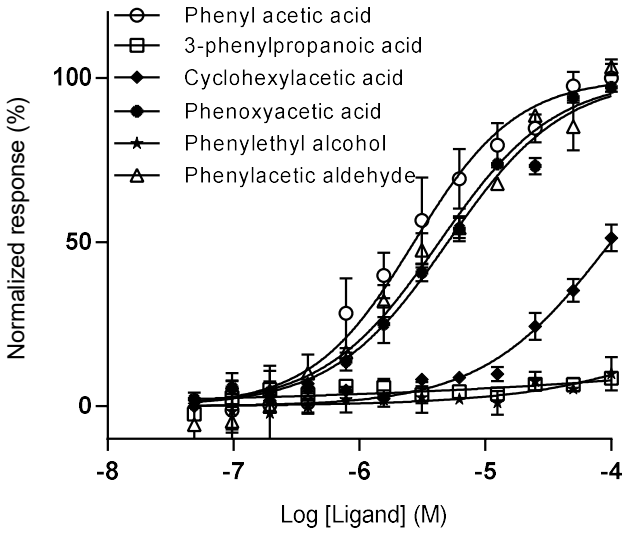
Dose-response analysis indicate that OR51L1 is very sensitive to this acid. Screening on a library of 41 carboxylic acids reveals a high specificity, with only the closely related cyclohexylacetic acid showing some activity at a high dose. The closely related phenyl propionic acid has a different odour (balsamic-cinnamic) and is completely inactive on this OR. Similarly, phenylethanol, a key rose-odorant, is inactive. Phenylacetic aldehyde is active, however, this is most likely due to the prolonged assay type of the luciferase assay, which allows for oxidation of aldehydes by cellular metabolism to happen as also observed for linear aldehydes activating OR specific for linear acids. Among the library of 438 commercial fragrance molecules, different esters of phenylacetic acid and phenoxyacetic acid activate this OR, which may be due to ester cleavage in the assay. As these esters retain the honey odor descriptor of phenylacetic acid, ester cleavage may also occur in the human nose. Based on these data, OR51L1 appears to be a specific OR for the detection of the typic sweet-honey odour of phenylacetic acid and it is a further example of a class I OR with a high specificity.



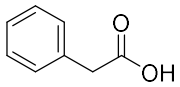
Screening of 77 class I OR variants containing the optimized C-terminal domain with 100 µM phenyl acetic acid



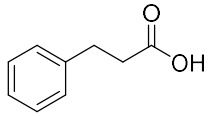
Screening of 42 carboxylic acids with the optimized OR51L1



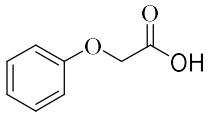
Activation of OR51L1 by odorants related to phenyl acetic acid



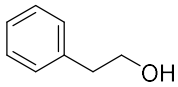
phenylacetic acid
honey, sweet, animalic
OTH: 0.18 ng/L
EC20% OR51L1: 5.5 x 10^{-7} M



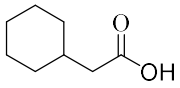
3-phenylpropanoic acid
balsamic, cinnamic
GTH: 0.25 ng
EC20% OR51L1: inactive



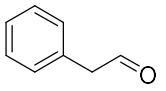
phenoxyacetic acid
EC20% OR51L1: 1.2 x 10^{-6} M



phenylethyl alcohol
floral, rosy
OTH: 4 ng/L
EC20% OR51L1: inactive



cyclohexylacetic acid
EC20% OR51L1: 1.9 x 10^{-5} M

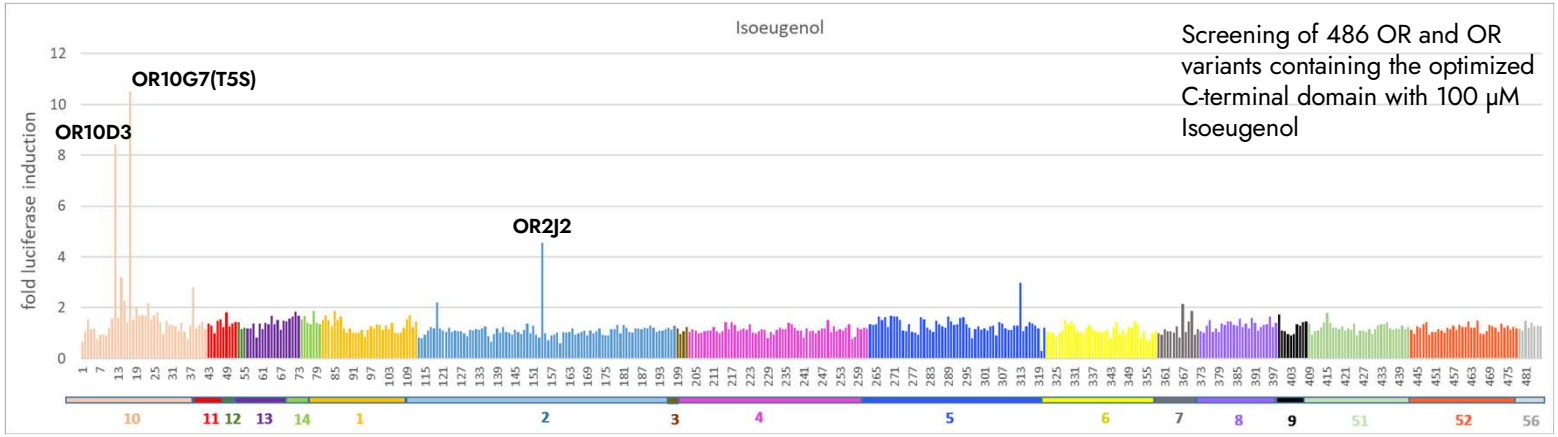


phenylacetic aldehyde
pungent-green, floral, hyacinth
OTH: 0.21 ng/L
EC20% OR51L1: 1 x 10^{-6} M

Givaudan

Structure, sensory data and OR activation of ligands tested on OR51L1

OR10D3 – an OR selective for Isoeugenol



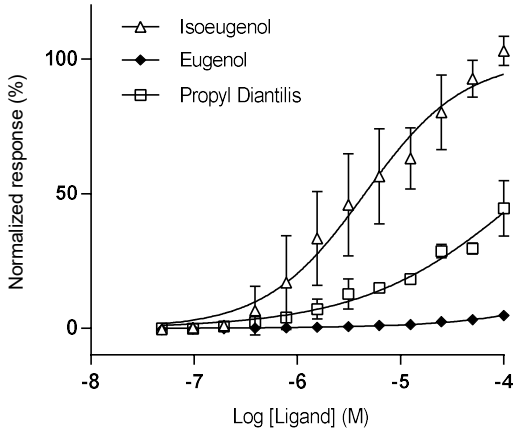
Screening of 486 OR and OR variants containing the optimized C-terminal domain with 100 µM Isoeugenol

Isoeugenol is a component of several essential oils such as clove oil, but it is also an important contributor to floral scents such as carnation or ylang-ylang. In this respect it is unique and cannot be replaced by the closely related eugenol, the main ingredient in clove. OR10G7 has been widely known as the OR responding to eugenol, and isoeugenol also activates this OR. However, in the screen on the OR library, OR10D3 was identified as a second OR strongly activated by isoeugenol.

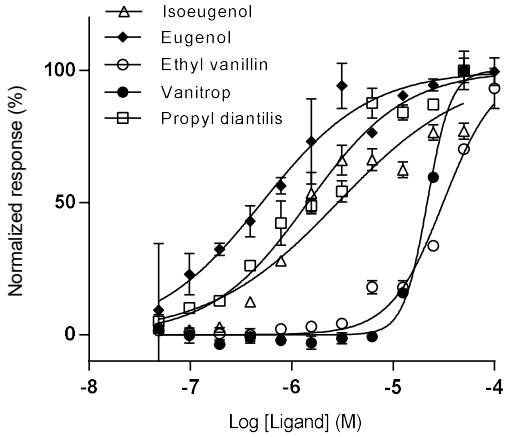
Among the library of 438 single fragrance ingredients, only isoeugenol and to a lesser extend propyl diantilis activated this OR.

It is noteworthy, that OR10D3 can selectively detect the difference of the double-bond position in isoeugenol, the only difference from eugenol. Interestingly, vanitrope is inactive on OR10D3 and only weakly active on OR10G7, despite close structural similarity to Isoeugenol.

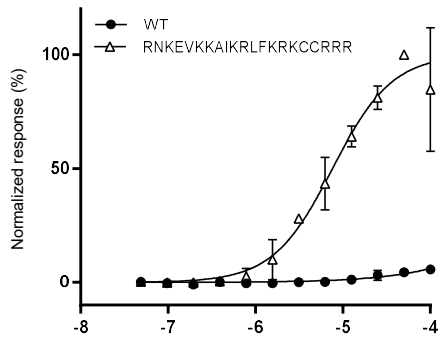
The wild-type OR10D3 is extremely weakly active in HEK293 cells, while the optimized variant is 2.5 orders of magnitude more sensitive.

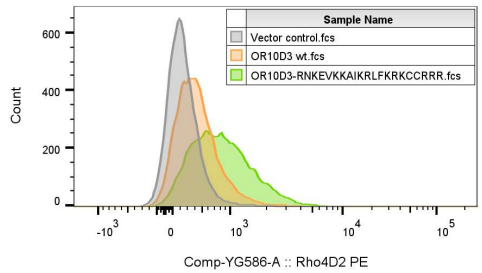


Activation of OR10D3 by spicy notes

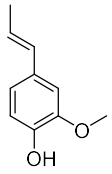


Activation of OR10G7 by spicy notes

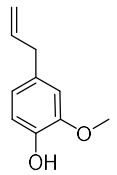




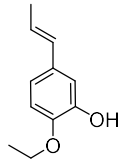
Activation and surface expression of OR10D3: Wildtype sequence compared to the optimized c-terminal sequence



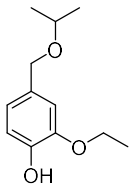
isoeugenol
spicy, floral
GTH: 0.15 ng
OTH: 0.04 ng/L
EC20% OR10D3: 9.8 x 10^{-7} M
EC20% OR10G7: 3.8 x 10^{-7} M



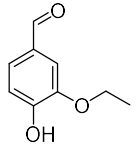
eugenol
spicy, metallic
OTH: 0.05 ng/L
EC20% OR10D3: inactive
EC20% OR10G7: 9.4 x 10^{-8} M



Vanitrope
balsamic, vanilla-like
OTH: 0.08 ng/L
EC20% OR10D3: inactive
EC20% OR10G7: 1.5 x 10^{-5} M



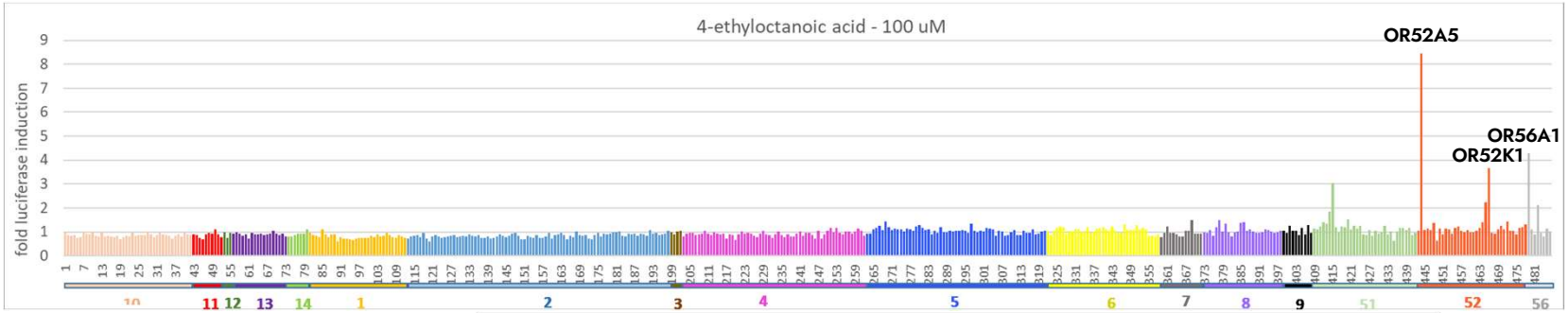
Propyl diantilis
spicy, vanilla-like, smokey
OTH: 0.8 ng/L
EC20% OR10D3: 1.3 x 10^{-5} M
EC20% OR10G7: 3.1 x 10^{-7} M

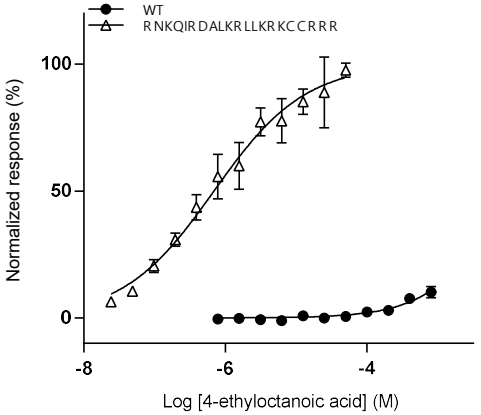


ethyl vanillin
balsamic, vanilla-like
GTH: 0.069 ng
OTH: 0.0097 ng/L
EC20% OR10D3: inactive
EC20% OR10G7: 1.4 x 10^{-5} M

Structure, sensory data and OR activation of ligands tested on OR10D3 and OR10G7

OR52A5 – the OR for goat acid





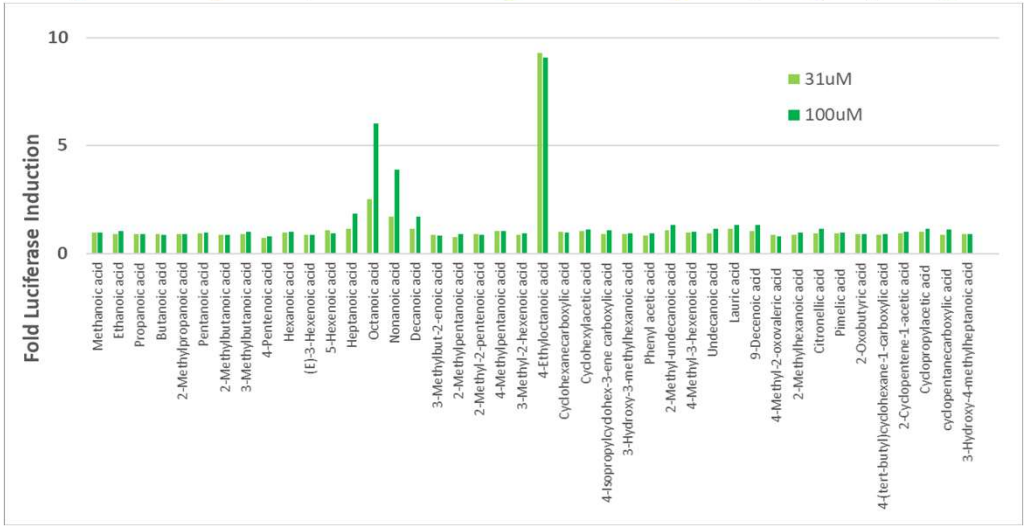
Activation of wild-type and optimized OR52A5

Goat acid is the key odorant in goat cheese with a uniquely low GTH. It most strongly activates OR52A5 among all 486 OR, with weak additional activity on class I OR56A1 and OR52K1, which are also responding to different straight C7 – C11 chain carboxylic acids.

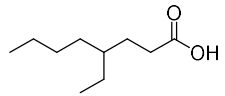
Among 42 carboxylic acids, a high specificity of OR52A5 was observed with only weak activity for octanoic acid and nonanoic acid.

In the library of 438 commercial fragrance molecules only very weak activity was observed for *p*-cresyl caprylate, an ester of octanoic acid, which is probably due to cleavage of the ester in the assay

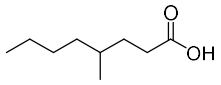
4-methyl-octanoic acid, which is also used in goat/mutton flavours is already 20-times weaker as compared to the ethyl-derivative and the straight octanoic acid is 400-fold less potent as compared to goat acid, confirming the high specificity of OR52A5 to goat acid.



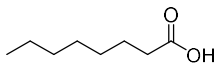
Screening of 42 carboxylic acids with the optimized OR52A5



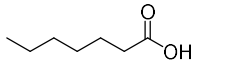
4-ethyl-octanoic acid
Goats, goat cheese, human sweat
GTH: 0.0005 ng
EC20% OR52A5: 5.8×10^{-8} M



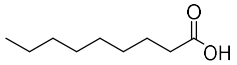
4-methyl-octanoic acid
used in goat and mutton flavors, human sweat
EC20% OR52A5: 1.2×10^{-6} M



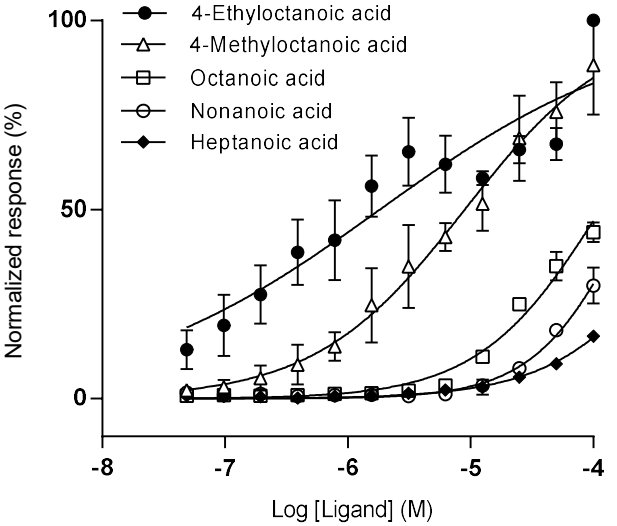
octanoic acid
(goat)milk
EC20% OR52A5: 2.4×10^{-5} M



heptanoic acid
EC20% OR52A5: 1.3×10^{-4} M



nonanoic acid
EC20% OR52A5: 6.0×10^{-5} M

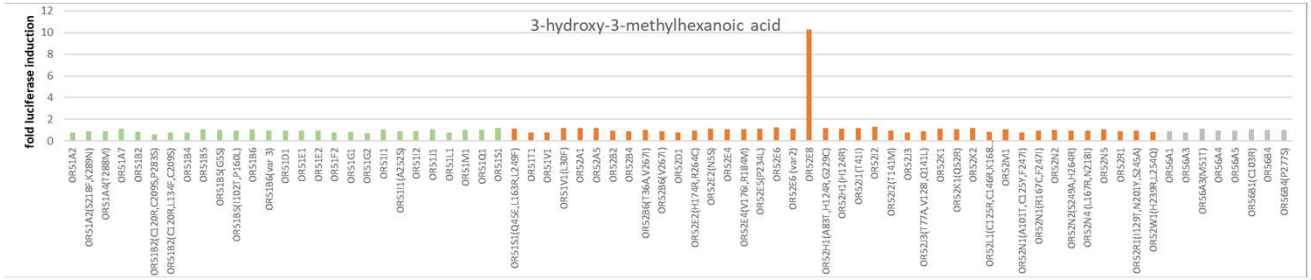


Activation or OR52A5 by carboxylic acids

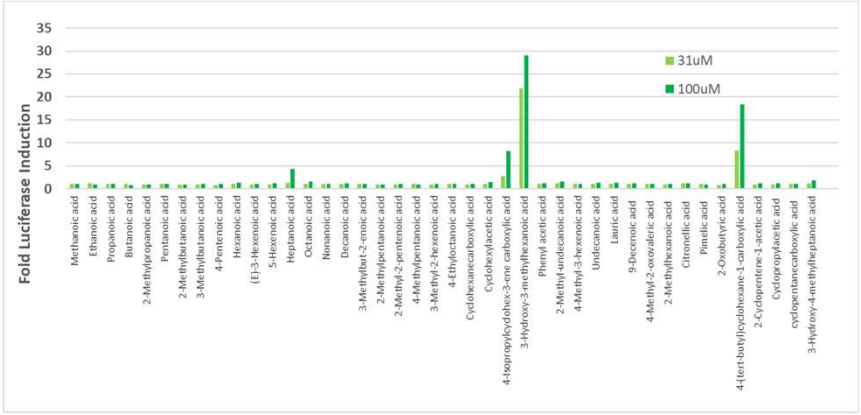
Structure, sensory data and OR activation of ligands tested on OR52A5

OR52E8 – an OR for the dominant human sweat odorant

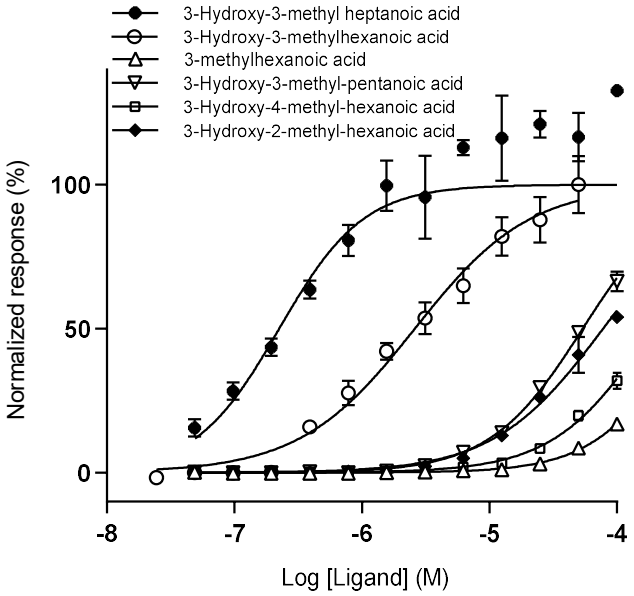
3-hydroxyl-3-methylhexanoic acid is the quantitatively dominant odorant acid in human axilla odor [1)2)]. It is detected by the human nose at very low concentration and therefore it has – next to some sulphur components - the strongest contribution to human axilla odor. Among the class I OR, only OR52E8 was activated by this acid, and no class II OR responded to this ligand (data not shown). On the other hand, screening 42 carboxylic acids with this OR, strong activation at 30 µM was only observed with this acid, showing a high selectivity for this OR to the specific human body odorant. Among related hydroxyl acids also occurring in human sweat [1)], only 3-hydroxyl-3-methylheptanoic acid is a strong ligand with an even higher potency. Comparison of the different hydroxy acids reveals a very high specificity of this OR: removal of either the methyl or the hydroxyl group leads to a > 100 fold lower activity. Similarly, shifting the position of the methyl group strongly reduces potency. Among 438 commercial fragrance ingredients, none of the compounds activated this OR, further confirming the high specificity to these specific acids. Activation for the wildtype OR52E8 had been reported before in a patent application [3)], however activation occurred in the 10^{-4} M range only, which could not yet explain the high sensitivity of this OR for this key human body odorant.



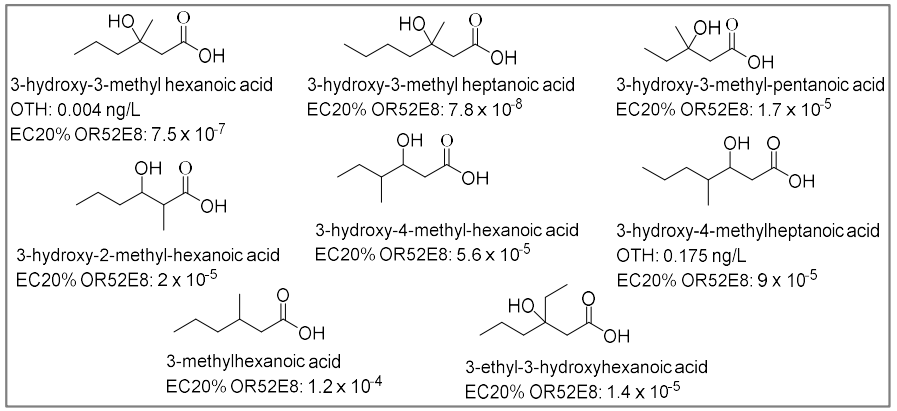
Screening of all class I OR with 3-hydroxyl-3-methylhexanoic acid



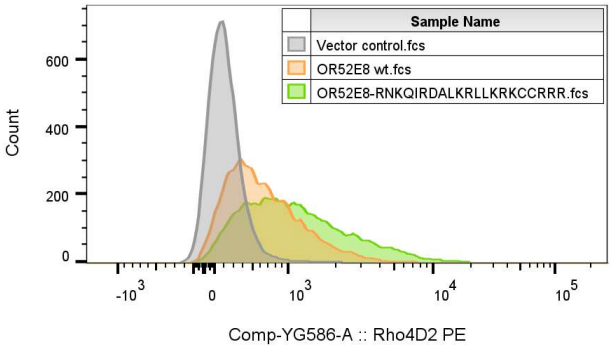
Screening a Library of carboxylic acids with OR52E8



Activation of OR52E8 by hydroxyacids



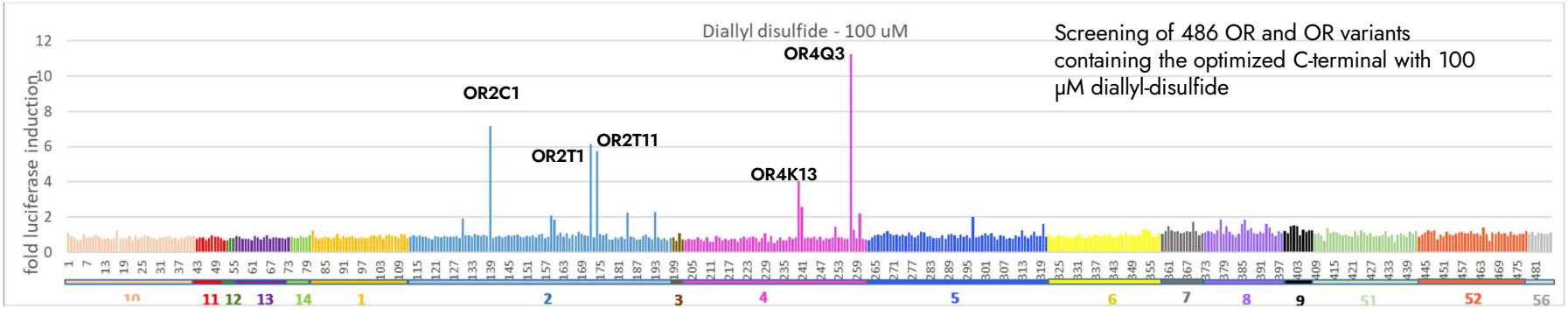
Structure and OR activation of ligands tested on OR52E8



Surface expression of wild-type and optimized OR52E8

1) Natsch, A., et al., A broad diversity of volatile carboxylic acids, released by a bacterial aminoacylase from axilla secretions, as candicate molecules for the determination of human-body odor type. Chemistry & Biodiversity, 2006. 3(1): p. 1-20.
2) Natsch, A. and R. Emter, The specific biochemistry of human axilla odour formation viewed in an evolutionary context. Philos Trans R Soc Lond B Biol Sci, 2020. 375(1800): p. 20190269.
3) Chatelain P, Veithen A. 2016 Olfactory receptors involved in the perception of sweat carboxylic acids and the use thereof. European patent application. EP3004157B1. Granted to ChemCom SA.

OR2T11 and OR2T1 – key OR for garlic disulfides

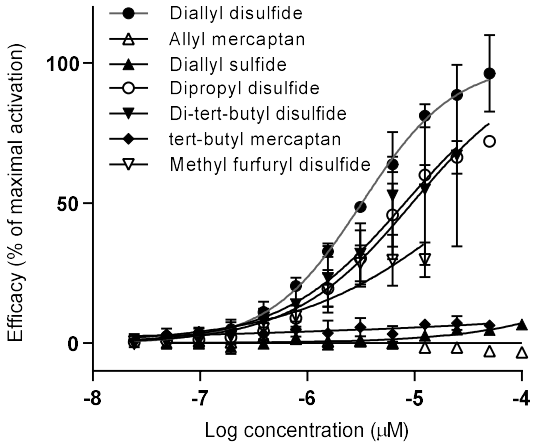


Diallyl disulfide is formed from the instable precursor allicin in garlic. It is the key aroma component of garlic next to other sulphur compounds such as diallyl trisulfid, diallyl sulfide, dipropyl disulfide and allyl mercaptan. Five OR were activated by diallyldisulfide when screened at 100 µM and in presence of 50 µM copper.

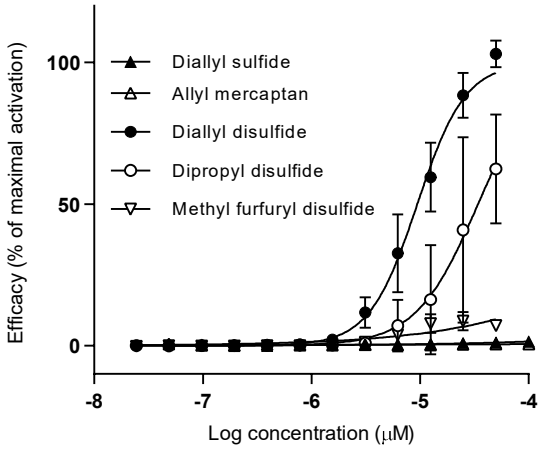
OR2C1 turned out to respond much more strongly to the monomer allyl mercaptan, rather than to diallyl disulfide in dose-response analysis. OR2C1 is highly sensitive to sulphur notes with a free SH groups, and is not responding to any of the 438 commercial fragrance ingredients.

Diallyl disulfide has the highest potency on OR2T11, and it is also the most potent ligand identified for this OR. Therefore, based on the current analysis, OR2T11 and to a lesser extend the closely related OR2T1 appear key OR to detect the garlic odour of diallyl disulphide. OR2T11 also responds to other dialkyl disulfides, but not to diallyl sulphide nor mercaptans, thus it appears selective to disulfides. OR2T11 had been reported to be strongly activated by tert-butyl mercaptan[1] used as natural gas odorant, however we cannot confirm this result when testing a fresh solution, while some activation is observed with aged solutions where the mercaptan may dimerize spontaneously, and indeed di-tertbutyl disulphide is a ligand of OR2T11.

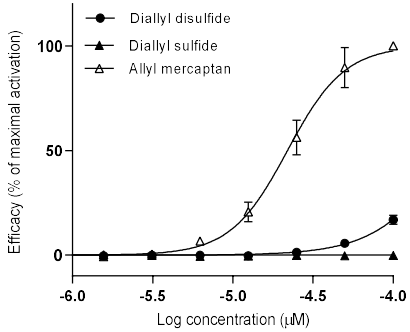
OR4Q3 is a rather broadly tuned OR responding to diallylsulfide only with an EC20 of 10^{-5} M and in parallel responding to an array of commercial fragrance ingredients, while OR4K13 is responding to diallylsulfide with an EC20 of 5 × 10^{-5} M and is also activated by non-sulphur compounds (data not shown).



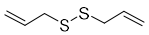
Activation of OR2T11 by garlic notes



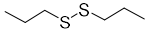
Activation of OR2T1 by garlic notes



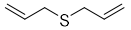
Activation of OR2C1 by garlic notes



diallyl disulfide
alliaceous, garlic, onion
EC20% OR2T11: 8.2×10^{-7} M
EC20% OR2T1: 6.1×10^{-6} M
EC20% OR2C1: 1.1×10^{-4} M



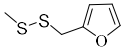
dipropyl disulfide
Garlic, green onion, sulfureous
EC20% OR2T11: 1.6×10^{-6} M
EC20% OR2T1: 1.4×10^{-5} M
EC20% OR2C1: n.d.



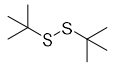
diallyl sulfide
sulfurous, onion, garlic, horseradish
EC20% OR2T11: 7×10^{-4} M
EC20% OR2T1: inactive
EC20% OR2C1: n.d.



allyl mercaptan
alliaceous, onion, garlic, leek
EC20% OR2T11: inactive
EC20% OR2T1: inactive
EC20% OR2C1: 1.2×10^{-5} M



methyl furfuryl disulfide
roasted coffee, toasted onion and garlic
EC20% OR2T11: 2.8×10^{-6} M
EC20% OR2T1: 2.7×10^{-4} M
EC20% OR2C1: n.d.



di-*tert*-butyl disulfide
garlic, green onion, sulfureous
EC20% OR2T11: 1.3×10^{-6} M
EC20% OR2T1: n.d.
EC20% OR2C1: n.d.



tert-butyl mercaptan
used as natural gas warning odorant
reported as OR2T11 ligand
only active on OR2T11 if aged solution is used

Ligands tested on OR2T1, OR2T11 and OR2C1

1) Li, S., Ahmed, L., Zhang, R., Pan, Y., Matsunami, H., Burger, J. L., Block, E., Batista, V. S.and Zhuang, H., (2016) Smelling Sulfur: Copper and Silver Regulate the Response of Human Odorant Receptor OR2T11 to Low-Molecular-Weight Thiols. J Am Chem Soc.10.1021/jacs.6b06983